Introduction to the Module

Friday, September 18, 2026 10:05 AM

Module Aims

This module covers the analysis and design of renewable and sustainable energy systems. It covers the various types of renewable energy and the resources available. It includes

- **Wind power:** wind probability distributions, wind turbine performance and control, comparison of generator types
- **Hydro and tidal power:** resource assessment, turbine types and principles
- **Solar power**, including PV cell equivalent circuit, analysis of losses, matching to DC and AC power systems
- **Wave power systems**, including wave energy characteristics, types of energy converter
- **Economic and environmental assessment** of energy conversion technologies.

Learning Outcomes

1. Evaluate different type of energy sources and understand their trends, which may include **conventional and non-conventional** options depending on the context and application
2. **Select appropriate** type of renewable energy source and energy storage to mitigate risks associated with security of electrical supply for a given application
3. Analyse various **energy storage systems** associated with renewable power generation.
4. Determine **high level system sizing and component** understanding for the renewable power generation.
5. **Demonstrate an understanding** of various renewable energy sources to design and analyse an **integrated renewable energy system**.
6. Analyse options (i.e. policies, advices or suggestions) aimed at **improving efficiency** in renewable energy generation and consumption.
7. Develop pragmatic knowledge of renewable energy sources and systems.

Learning Activities

Week	Topics	
1	Intro to the Module	Dr. Freddy
2	Efficiency Policy	
3	Solar Cells	
4	Grid-Connected Photovoltaic System Design	
5	Safety Standards, Guidance and Regulation	
6	Energy Storage	
7	Fossil Fuel and Carbon Emission	Prof. Yousif
8	Wind Power	
9	Hydro Power	
10	Ocean Energy, Tidal Power and Wave Energy	
11	Geothermal Energy	
12	Revision	

Delivery Method

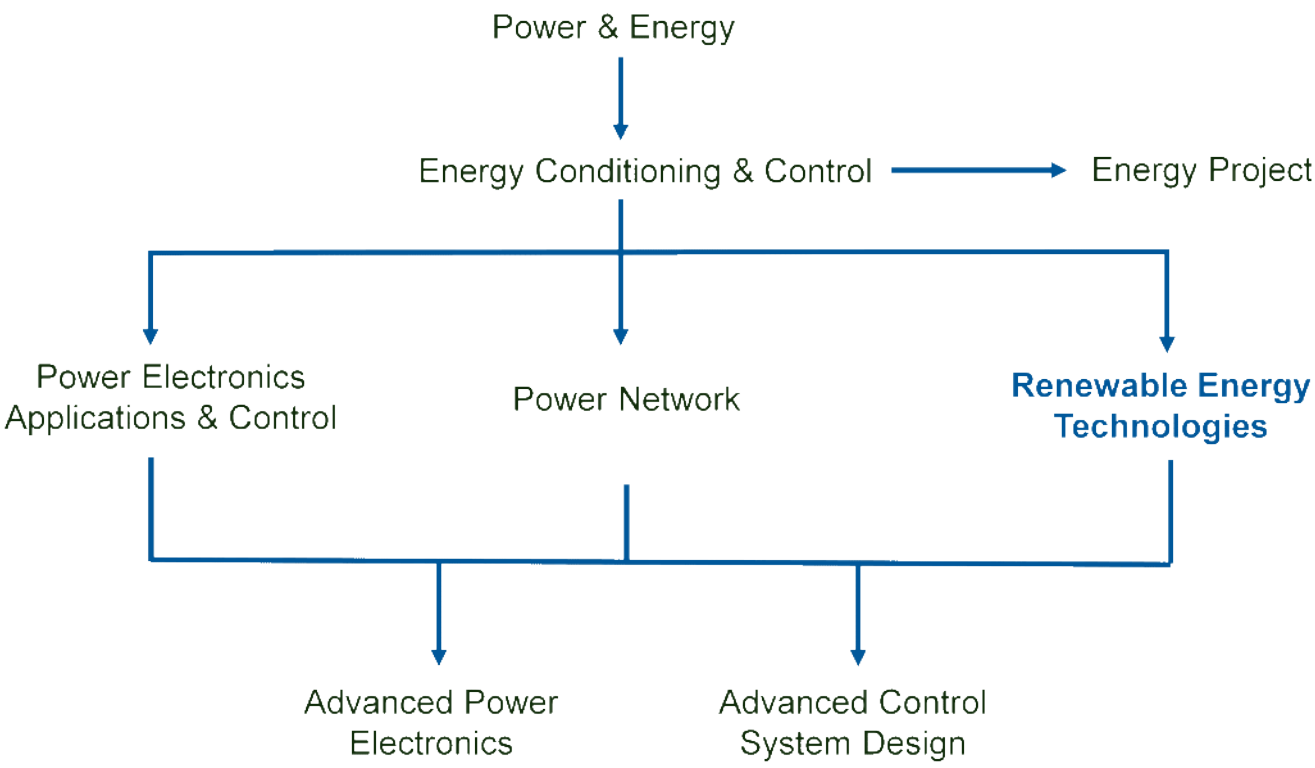
Modes	Number of Sessions	Number of Hours	Duration
Lecture	1 session/week	2 hours/session	12 weeks
Tutorial	1 session/week	1 hour/session	12 weeks

Assessment

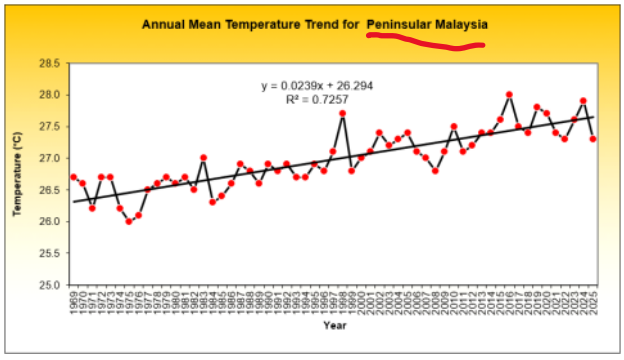
Assessment	Weightage	Requirements
Coursework 1	25%	
Coursework 2	25%	
Exam	50%	2 hours

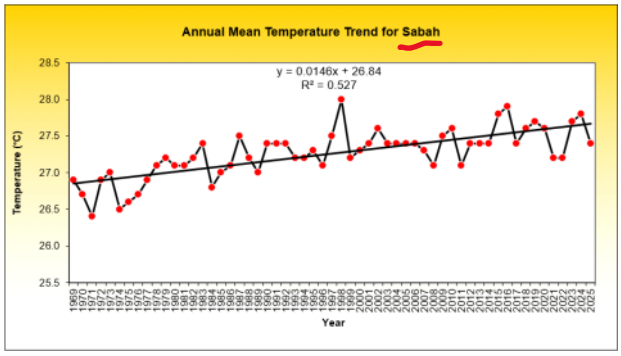
Link to Other Module

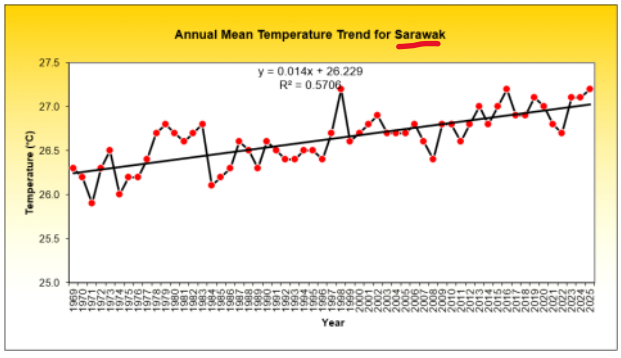
Friday, September 18, 2026 11:01 AM



Climate Change



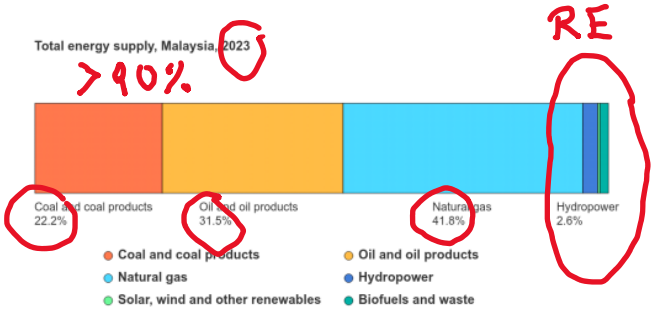




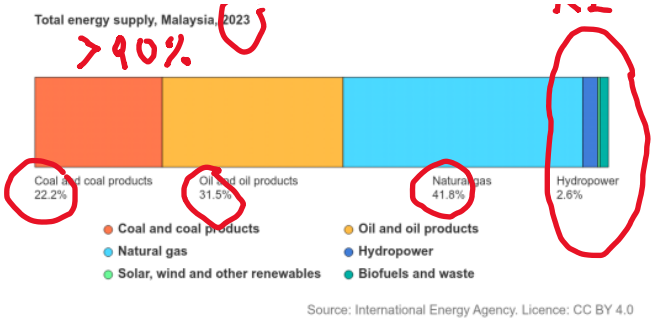
- Rapid economic and infrastructure development drives climate change primarily by increasing greenhouse gas emissions from fossil fuel combustion, urban expansion, and heavy construction.
- Extreme climate change will lead to extreme heat, sea level rise, and sea ice free arctic.
- Currently, our planet is already about 1.44°C warmer than it was in the late 1800s.
- The temperature rise in Peninsular is the greatest, with about 1 degree celcius since 1969.

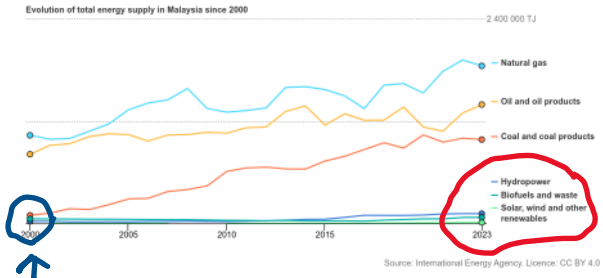
Source: Malaysian Meteorological Department https://www.met.gov.my/en/pendidikan/perubahan-iklim-and-kesan-rumah-hijau/

Energy supply



- Heavily reliant on **fossil fuels**, which account for roughly **95%** of total primary energy.
- Renewable Energy accounts only about **4%** of the energy supply.





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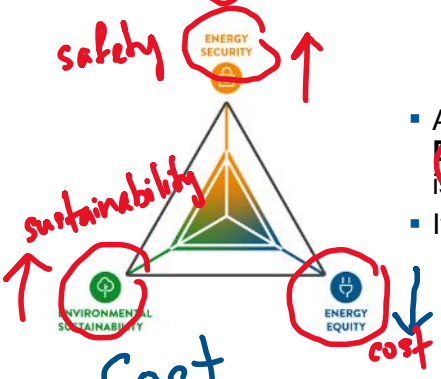
Climate Change Initiatives

- Paris Agreement - limit global warming to well **below 2°C**, preferably to 1.5°C, compared to pre-industrial levels.
- The United Nations Framework Convention on Climate Change (UNFCCC) - SDG 13 - investment in renewable energy

Energy Trilemma

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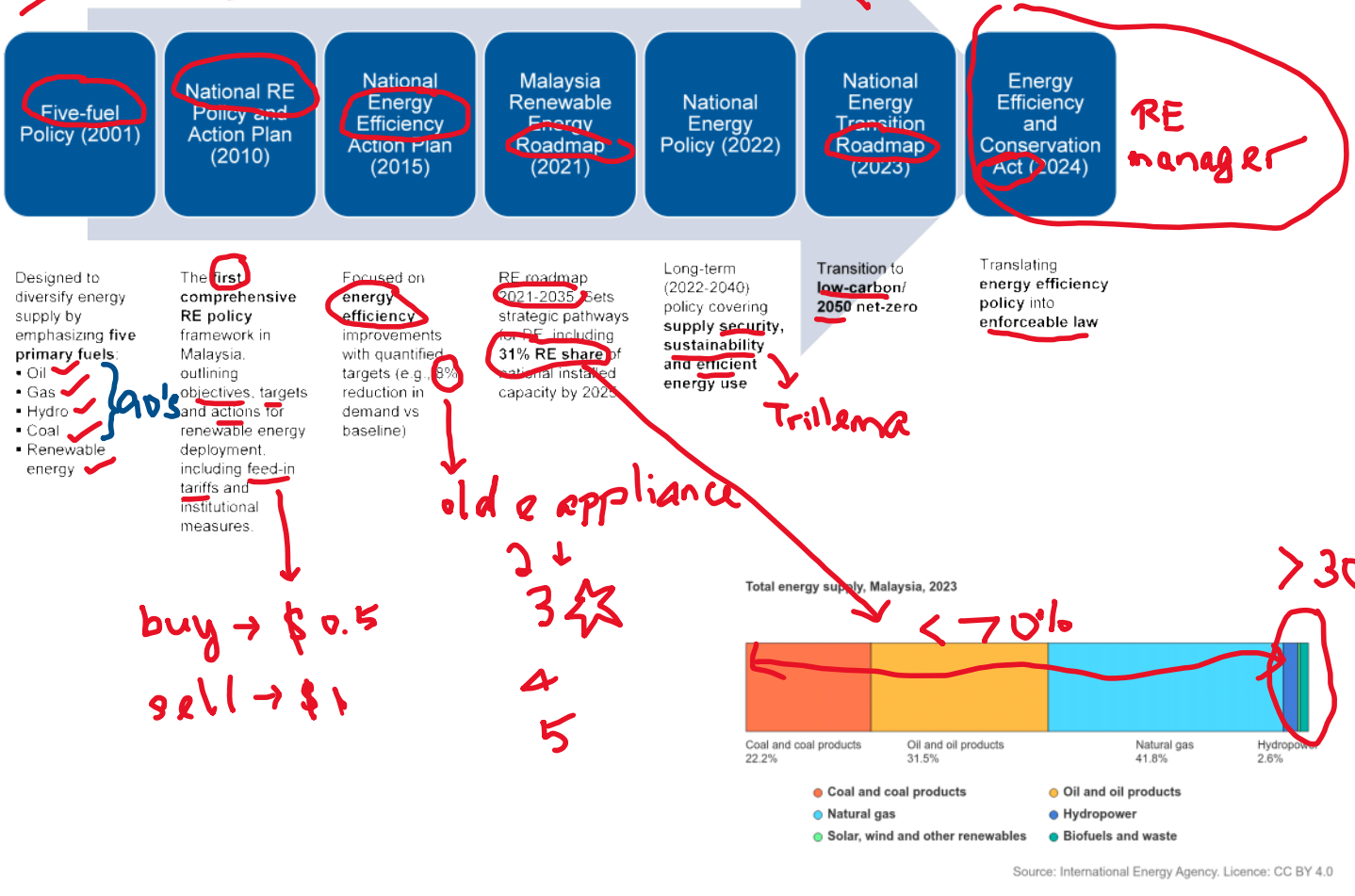
Energy Trilemma



- According to the World Energy Council, the 'trilemma' between **Energy Equity**, **Environmental Sustainability** and **Energy Security** is one of the biggest challenges of this transition.
- It is difficult to emphasize one without compromising the other.

- **Energy Equity** of this transition - Ability to provide universal access to affordable, fairly priced and abundant energy for domestic and commercial use.
- **Environmental Sustainability** - Transition of energy system towards mitigating and avoiding potential environment and climate change impacts.
- **Energy Security** - Capacity to meet current and future energy demand reliably, withstand and recover swiftly from system shocks with minimal disruption to supplies.

Timeline



Source: International Energy Agency. Licence: CC BY 4.0

Introduction

- Came into force in 2025 as Malaysia's first dedicated energy efficiency law.
- Introduces mandatory energy efficiency standards for industries, buildings, and large energy users (replaced Efficient Management of Electrical Energy Regulations 2008).

Energy Consumers – Industrial Sectors

- **Industrial facilities (Energy Consumers)** are regulated if their energy use meets or exceeds a threshold of **21,600 gigajoules (GJ) per year**. It is required to:
  - **Appoint a Registered Energy Manager (REM)** who is responsible for implementing and monitoring energy management systems, preparing energy efficiency reports, and advising on energy-saving measures.
  - **Establish and operate an Energy Management System (EnMS)** to systematically manage energy consumption.
  - **Conduct periodic energy audits** to identify inefficiencies and opportunities for savings.
  - **Submit Annual Energy Efficiency and Conservation Reports** to the Energy Commission, detailing energy consumption, management systems, and proposed improvements.

Office Buildings

- **For office buildings and other commercial buildings** that have a gross floor area (GFA) of **8,000 m² (about 86,000 sq ft) or more**:
- The Person in Charge (owner or building manager) **must apply for and display a Building Energy Intensity (BEI)** label at a conspicuous part of the building
- The energy intensity performance of the building must meet prescribed standards with **at least a two-star rating**. Non-compliance will require the building to undergo an energy audit and implement an energy efficiency improvement plan.
- The BEI is a benchmarking tool to monitor building energy performance by indicating the intensity of energy used per meter square area of the building.
- BEI is calculated by the ratio between annual energy consumption of a building (kWh/year) and building nett floor area (NFA)

$$BEI = \frac{Annual\ energy\ consumption\ (kWh)}{NFA\ (m^2)}$$

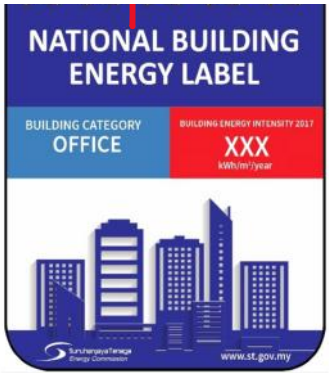
Building Energy Intensity Label



STAR RATING	BEI RANGE VALUE
5-Star	BEI ≤ 100

5 star: Most efficient

Building Category:
Office/ Hospital/ University/ School & others



Building energy performance Unit:
kWh/m²/year

STAR RATING	BEI RANGE VALUE
5-Star	BEI ≤ 100
4-Star	100 < BEI ≤ 130
3-Star	130 < BEI ≤ 160
2-Star	160 < BEI ≤ 250
1-Star	BEI > 250

Pass

Fail

Appliance Manufacturers and Importers

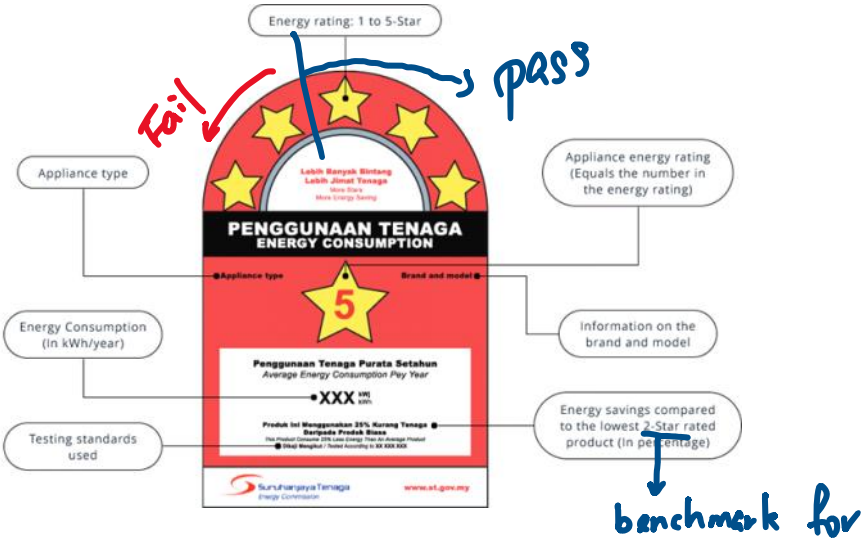
- **Appliance manufacturers and importers** are regulated through product efficiency controls.
- The regulated electrical appliances includes air conditioners, refrigerators & freezers, fans, lighting products, televisions, washing machines, microwave, and others
- EECA requirements are:
  - Manufacturers and importers are required to obtain a certificate of registration (COR)
  - Energy using appliances to meet the prescribed minimum requirement and obtain a Certificate of Efficiency (COE).
  - Energy using appliances to be affixed with an energy efficiency rating label that will also be issued together with COE.

Mandatory Minimum Energy Performance Standards (MEPS)

- Every regulated appliance must meet minimum efficiency levels set by the Energy Commission (ST).

EC

- MEPS minimum requirement is **2-Star.**
- Products that fail MEPS:
  - Cannot be manufactured for sale
  - Cannot be imported
  - Cannot be distributed or sold

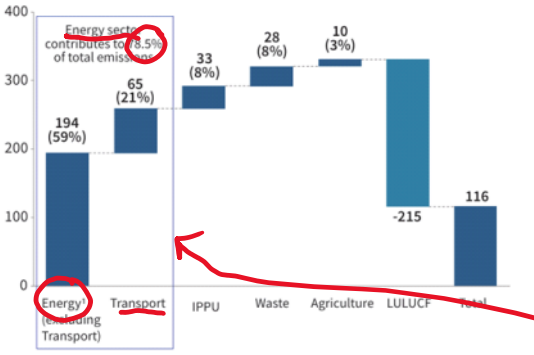


energy saving

- Air Conditioner
  - Type :Non-ducted Single Split Wall Mounted
  - Capacity ≤ 25,000 btu/hr
- Fan
  - Ceiling Fan with diameter less than 60 inch
  - Wall fan,desk fan,table fan with diameter less than 16 inch
- Refrigerator
  - 1-door & 2-door only
- Television
  - Type :LCD,PLASMA,LED,CRT
  - Screen size up to or equal to 70 inch
- Lamp
  - T5 & T8 Fluorescent Lamp
  - Self ballasted single capped CFL
  - Single Capped Fluorescent Lamp & Circular Fluorescent Lamp
  - Self ballasted LED Lamp

Introduction

- In 2019, the **energy sector** emitted 259,326.11 Gigagram CO2 equivalent (GgCO2eq), or 78.5% of total emissions. This was followed by industrial processes and product use (IPPU) at 10%, waste at 9% and agriculture at 3%.



1 Refers to emissions from energy industries, manufacturing industries and construction, other sectors and non-specified energy emissions, and fugitive emissions from fuels.

Source: Malaysia's Fourth Biennial Update Report submitted to the UNFCCC (2022)

combat

National Energy Policy (NEP)

- NEP provides a long-term, integrated direction for Malaysia's **energy sector** up to 2040, while supporting the country's energy transition and climate commitments.
- The policy aligns energy planning with economic development, environmental sustainability, and social well-being, and serves as the umbrella under which roadmaps such as **MyRER** and **NETR** operate.
- NEP has three main objectives (balance energy trilemma):
  - enhancing macroeconomic resilience and energy security (safety)
  - achieving social equitability and affordability (cost)
  - ensuring environmental sustainability (sustainability)

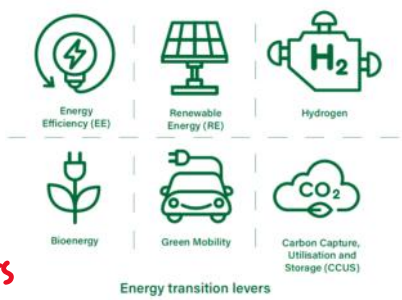
Selected Targets	2018	Low Carbon Nation Aspiration 2040
1. Percentage of urban public transport modal share	20%	50%
2. Percentage of electric vehicle (EV) share	<1%	38%
3. Alternative fuel standard for heavy transport	B5	B30
4. Percentage of Liquefied Natural Gas (LNG) as alternative fuel for marine transport	0%	25%
5. Percentage of industrial and commercial energy efficiency savings	<1%	11%
6. Percentage of residential energy efficiency savings	<1%	10%
7. Total installed capacity of RE	7,597 MW	18,431 MW
8. Percentage of coal in installed capacity	31.4%	18.6%
9. Percentage of RE in TPES	7.2%	17%

National Energy Transition Roadmap (NETR)

- NETR is national roadmap to achieve net-zero GHG emission by 2050.
- It provides strategic plan to shift energy systems from fossil fuel-based to greener and low-carbon systems.
- NETR aims to lower the energy sector's greenhouse gas emissions by **32% by 2050**, relative to the 2019 baseline.
- NETR identifies 10 flagship catalyst projects under six key levers, including
  - Energy efficiency ✓
  - Renewable energy ✓
  - Hydrogen ✓



- Energy efficiency
- Renewable energy
- Hydrogen
- Bioenergy
- Green mobility
- Carbon capture, utilization and storage



Energy sector → industries, transports

Overview of NEP and NETR Goals

Sector and Key Driver	2040 DTN Low Carbon Nation Asp.	2050 NETR Responsible Transition
EE Energy Efficiency – Industry and Commercial energy efficiency savings (%)	11%	23%
EE Energy Efficiency – Residential energy efficiency savings (%)	10%	20%
RE Renewable energy – Coal share of installed capacity (%)	19%	0%
RE Renewable energy – RE share of installed capacity (%)	41%	70%
HY Hydrogen – Green hydrogen production (MTPA)	N/A	Up to 2.5 MTPA
HY Hydrogen – Grey hydrogen feedstock phase-off (%)	N/A	100%
HY Hydrogen – Hydrogen hubs (#)	N/A	3
BI Bioenergy – Biofuel capacity (billion litres)	N/A	3.5
BI Bioenergy – Bioenergy power generation (GW)	N/A	1.4
GM Green Mobility – Land – Urban public transport modal share (%)	50%	60%
GM Green Mobility – Land – xEV (4W) share of fleet (%)	38%	80%
GM Green Mobility – Land – E2W share of fleet (%)	N/A	80%
GM Green Mobility – Land – Light vehicle fuel economy	N/A	~30%
GM Green Mobility – Land – Heavy transport fuel economy	N/A	~24%
GM Green Mobility – Land – Biofuel blending for heavy transport (%)	B30	B30
GM Green Mobility – Land – Hydrogen penetration for heavy transport (%)	N/A	5%
GM Green Mobility – Marine – LNG penetration as alternative fuel in marine transport (%)	25%	N/A
GM Green Mobility – Marine – Green fuel penetration in marine transport (%)	N/A	40%
GM Green Mobility – Air – SAF blending mandate by 2050 (%)	N/A	47%
CC CCUS – Number of CCUS clusters (#)	N/A	3-6
CC CCUS – CO_2 storage capacity (Mtpa)	N/A	40-80

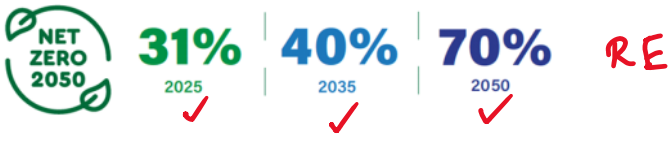
more ambitious & comprehensive

	2030	2040	2050
Energy Efficiency	~RM2 b in smart devices installation; ~RM2 b into retrofitting gov. buildings	~RM6 b in smart devices installation; ~RM7 b into retrofitting gov. buildings	~RM7 b in smart devices installation
Renewable energy	~7GW of solar PV (31% RE of installed capacity)	~27GW of solar PV (52% RE of installed capacity)	~57GW of solar PV (70% RE of installed capacity)
Hydrogen	Establishment of 1 hydrogen hub	Establishment of 2 additional hubs	Up to 2.5MTPA local green H_2 production
Bioenergy	0.6 billion litres of biofuel production; ~500 MW of bioenergy installed capacity	1.8 billion litres of biofuel production; ~600 MW of bioenergy installed capacity	3.5 billion litres of biofuel production; ~1,400 MW of bioenergy installed capacity
Green Mobility	20% of 4W fleet as xEVs; 20% of 2W fleet electrified	50% of 4W fleet as xEVs; 50% of 2W fleet electrified	80% of 4W fleet as xEVs; 80% of 2W fleet electrified
CCUS	3 CCUS hubs; Up to 15 MTPA of CO_2 storage capacity	1 carbon capture hub; 40 MTPA of CO_2 storage capacity	2 additional carbon capture hubs; 80 MTPA of CO_2 storage capacity

Malaysia Renewable Energy Roadmap (MyRER)

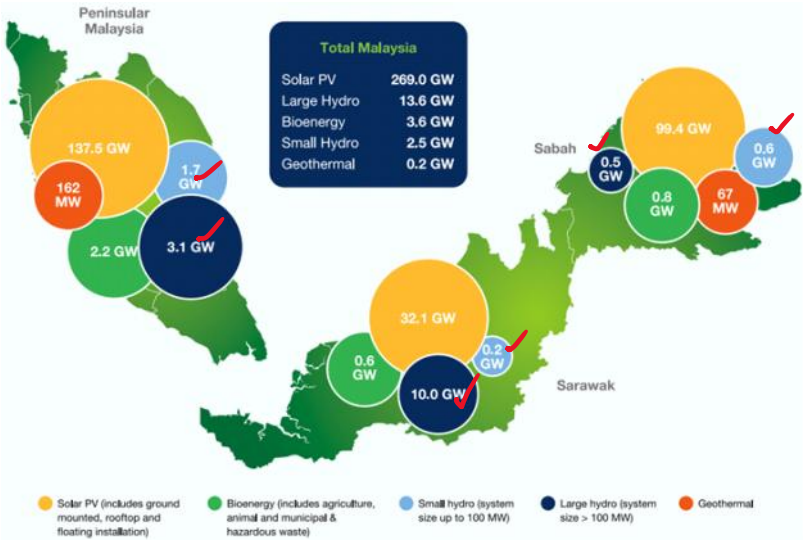
regulated by → SEDA Malaysia

- The primary goal of MyRER is to increase the share of renewable energy in Malaysia's power capacity mix. The specific targets are:
  - **31% share of RE** in the national installed capacity mix by **2025**.
  - **40% share of RE** in the national installed capacity mix by **2035**.
  - **70% share of RE** in the national installed capacity mix by **2050 (NETR)**.



RE

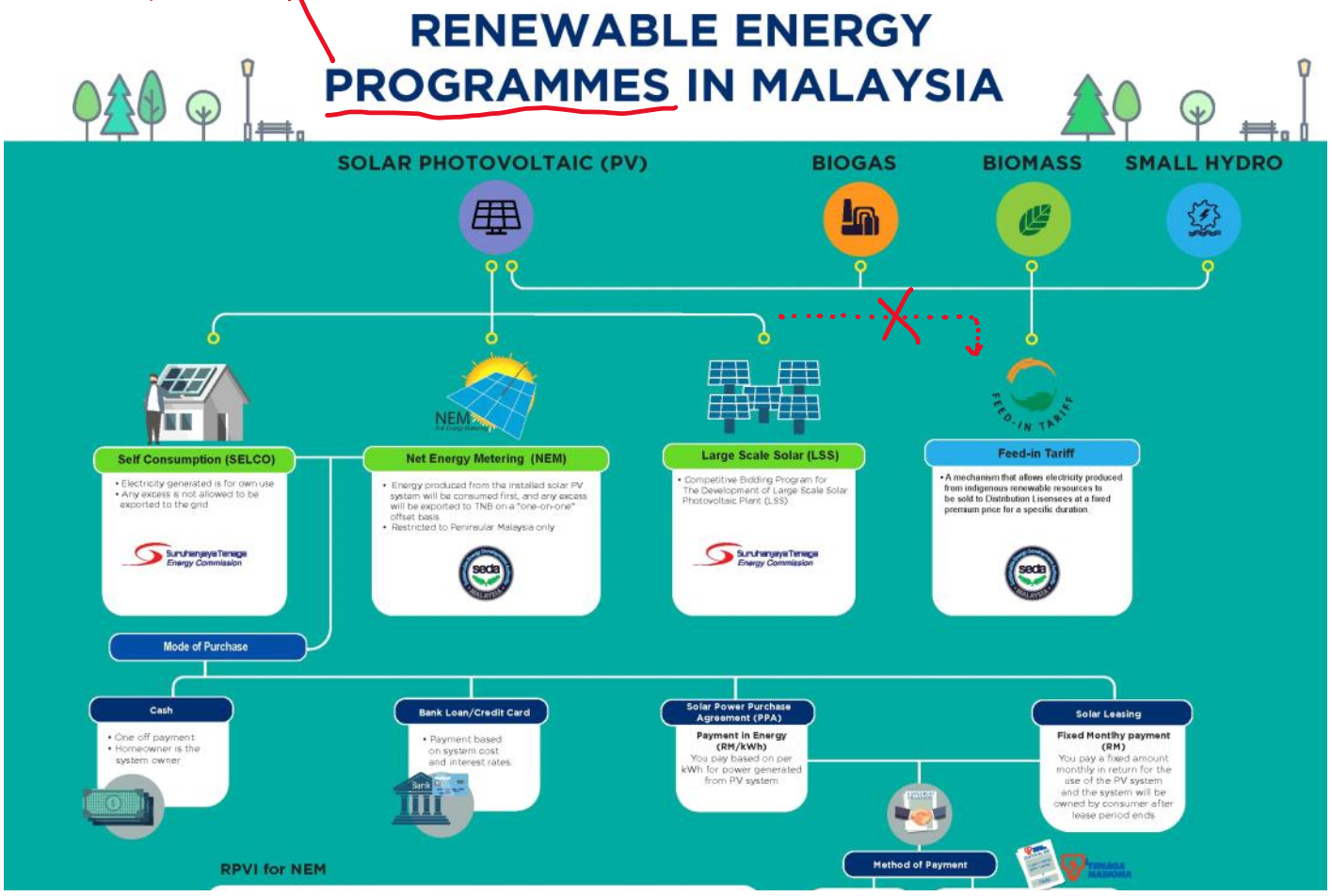
- Malaysia has high potential for RE, due to **year-round solar irradiation, abundant biomass resources, extensive river basins**





- This represents a projected increase in RE capacity of **17,996 MW** by **2035**.

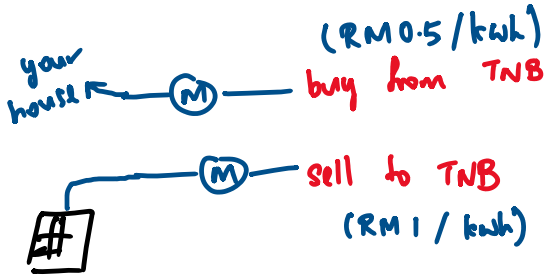
encourage installations increase RE share



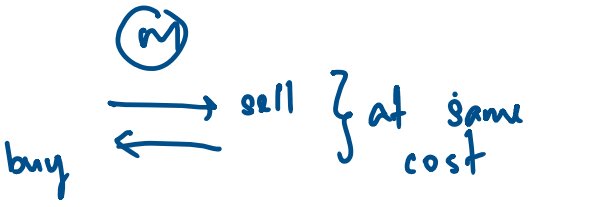


Source: https://www.seda.gov.my/about-seda/re-programmes/

FiT (2 meters)



NEM (1 meter)



use first the solar consumption
→ extra → sell to TNB

Solar Cells

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PV → Photovoltaic

Solar

Thermal

Photovoltaic (PV)

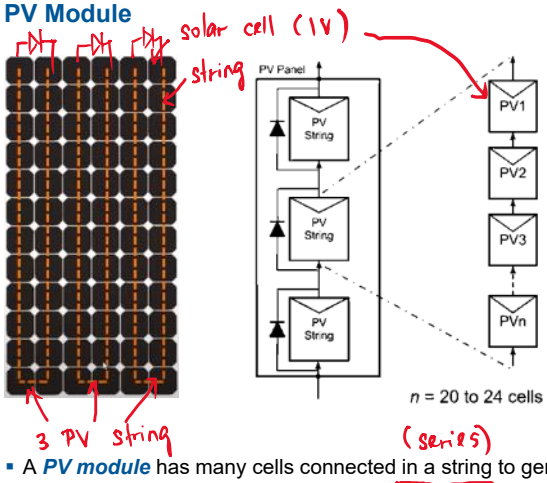
Light → Heat

Light → Electricity

Thermal VS Photovoltaic



PV Module



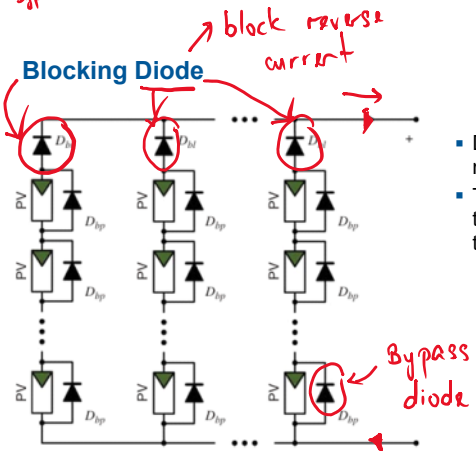
- A ***PV module*** has many cells connected in a string to generate higher voltage.
- The electrical characteristics of each PV cell is different from others due to manufacturing tolerance.

Bypass Diode



- A ***bypass diode*** is mounted into a junction box on the rear side of the panel and is usually inactive.
- It is connected (in parallel) across PV sub-string (module).

Blocking Diode



- During low solar irradiance condition or at night, the PV modules stop producing the power and become idle.
- To avoid the flow of current from load (battery/grid/etc) to PV modules, a diode, called ***blocking diode*** is used to block the current flow.

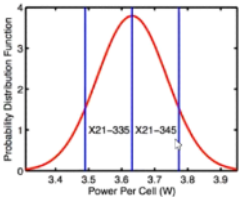
Power Mismatch

- PV module mismatch, causing significant power losses (often ~10%), occurs when interconnected solar panels operate at different currents or voltages.
- Major causes include
  - manufacturing tolerance variations ✓
  - partial shading (leaves, dirt, structures) ✓
  - uneven aging/degradation of modules ✓
  - temperature differences ✓
  - mismatched panel specifications (mixing different wattages) ✓

Power Mismatch – Manufacturing



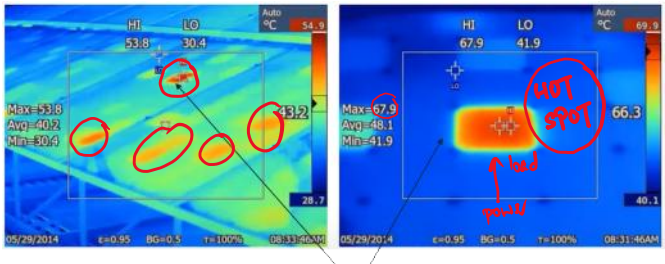
Binning is the process of categorising individual PV cells into groups (bins) based on similar electrical performance characteristics to minimise power losses when assembling them into a solar module



Power Mismatch – Partial Shading

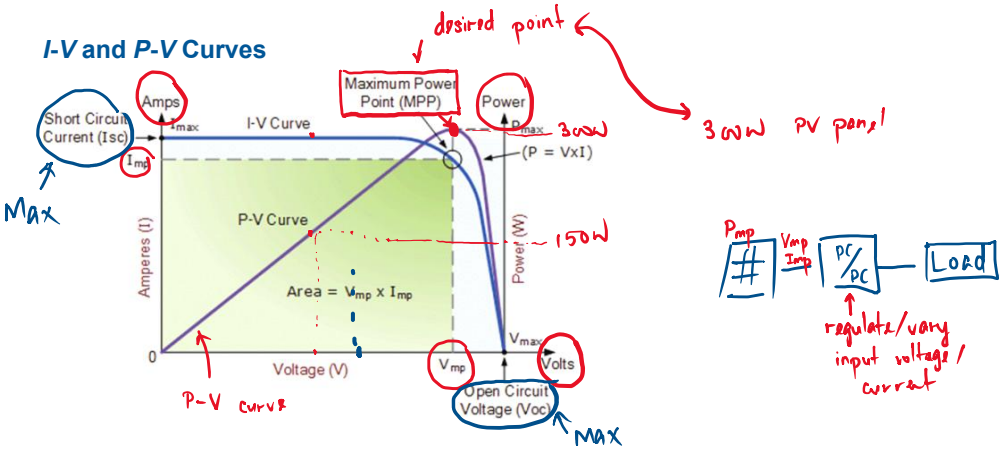


Partial shading occurs due to bird drops, shadowing, dust, etc which lead to mismatch losses.



The power from neighbouring PV cells is absorbed by the shaded cell and it becomes **HOT**.

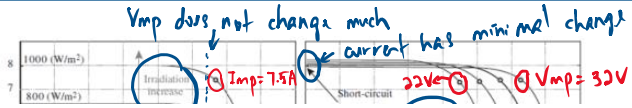
I-V and *P-V* Curves

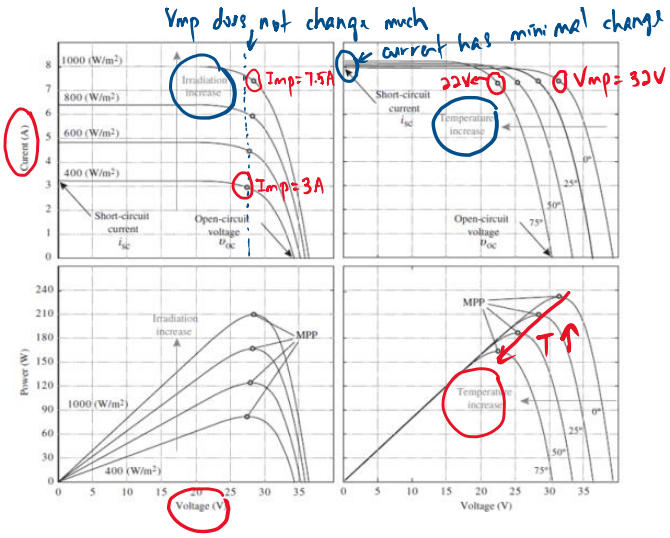


- The **temperature** has a significant effect on the **open-circuit voltage value**.
- The **short-circuit current** is linearly dependent on the **irradiance (G)**.
- The cell temperature (T), is calculated from the ambient temperature T_a by using the nominal operating cell temperature ($NOCT$):

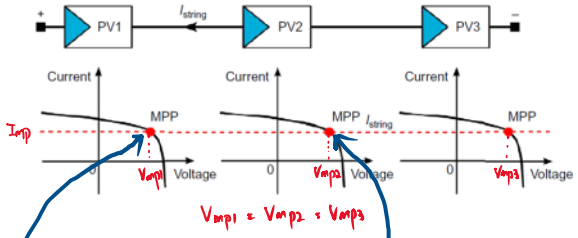
$$T = T_a + \frac{NOCT - 20}{800} \cdot G$$

NOCT is defined as the temperature reached by open-circuit cells in a module under the following conditions: irradiance on cell surface G = 800 W/m^2, air temperature T_a = 20°C, wind velocity = 1 m/s, mounting = open back side

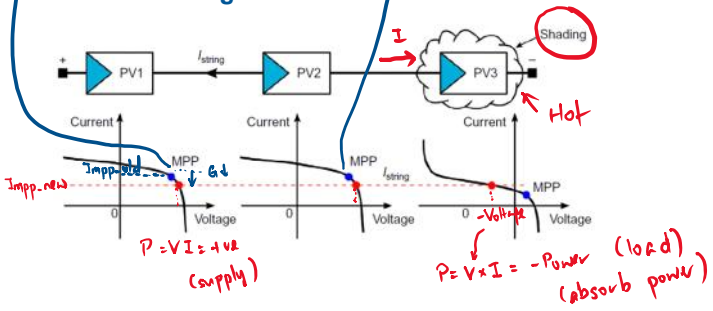


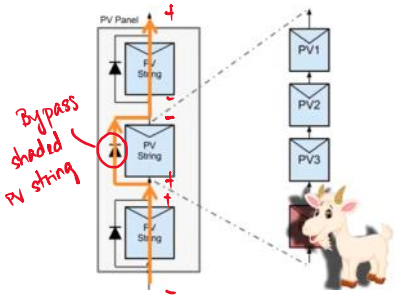


Normal Condition



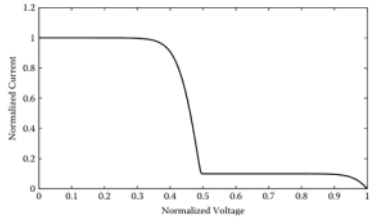
Partial Shading Condition

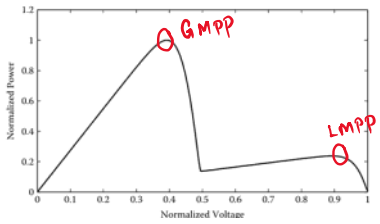




- Once that reverse voltage exceeds the diode's forward threshold (~0.5–0.7 V), the **bypass diode conducts**.
- The shaded sub-string is bypassed, protecting the cells from damage.





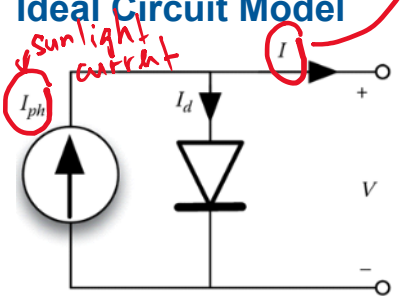


- Without bypass diodes, the shaded cells would be forced into reverse bias, causing hot-spots and large power loss.
- With bypass diodes, it bypasses the shaded module when the sub-string voltage becomes negative. That shaded section is bypassed, current can continue to flow through the rest of the module.
- However, this generates **multiple peaks** in *I-V* and *P-V* curves which requires complicated advanced techniques to harness the maximum power.

Equivalent Circuit

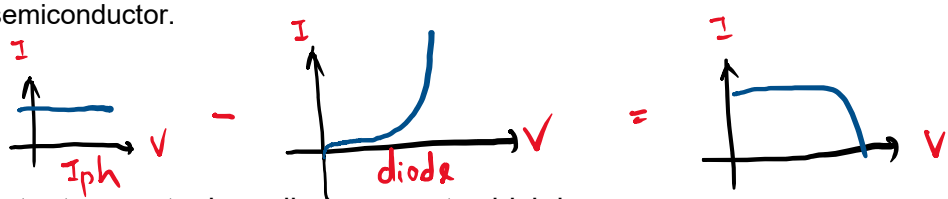
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Ideal Circuit Model



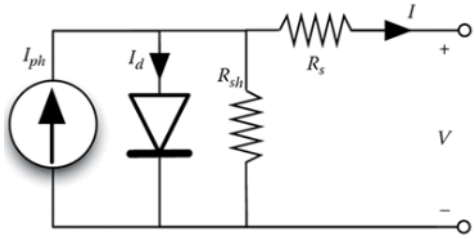
Ideal equivalent circuit of a PV module

- **Photocurrent source (I_{ph})** represents the current generated by sunlight (proportional to irradiance).
- **Diode (D)** represents the p–n junction of the PV cell. It accounts for the exponential *I–V* relationship of the semiconductor.



- The *I-V* curve is similar to a constant current minus diode current, which is nonlinearly dependent on its voltage.
- Therefore, an ideal equivalent circuit is a current source in parallel with a diode.

Single Diode Model with Losses



Single-diode model

- **Series resistance (R_s)** represents internal resistances (contacts, metal interconnections). Higher R_s reduces output voltage and power.
- **Shunt resistance (R_{sh})** represents leakage paths across the junction. A high R_{sh} is desirable (minimizes leakage losses).

- The single-diode model is the most widely used because it balances accuracy with simplicity.
- It represents the PV cell (or module) using standard electrical components that mimic the physical effects inside the solar cell.

- The output current I of the PV cell is given by:

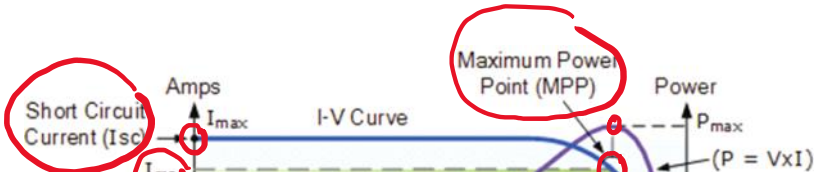
$$I = I_{ph} - I_0\left(e^{\frac{q(V+IR_s)}{nkT}} - 1\right) - \frac{V + IR_s}{R_{sh}}$$

- If $R_s \to 0$ and $R_{sh} \to \infty$, the equation simplifies to:

$$I = I_{ph} - I_0\left(e^{\frac{qV}{nkT}} - 1\right)$$

I = output current (A)
V = output voltage (V)
I_{ph} = photocurrent
I_0 = diode reverse saturation current
q = electronic charge (1.602×10^{-19} C)
k = Boltzmann constant (1.381×10^{-23} J/K)
T = cell temperature in Kelvin
n = diode ideality factor (1–2)
R_s = series resistance (Ω)
R_{sh} = shunt resistance (Ω)

- Short-Circuit Current (I_{sc}): At V = 0
  $I_{sc} \approx I_{ph}$
- Open-Circuit Voltage (V_{oc}): At I = 0



$I_{sc} \approx I_{ph}$

- Open-Circuit Voltage (V_{oc}): At I = 0

$$V_{oc} = \frac{nkT}{q} \ln\left(\frac{I_{ph}}{I_0} + 1\right)$$

- Maximum Power Point (MPP):

$$P_{mpp} = V_{mpp} \times I_{mpp}$$

- Fill factor (FF):

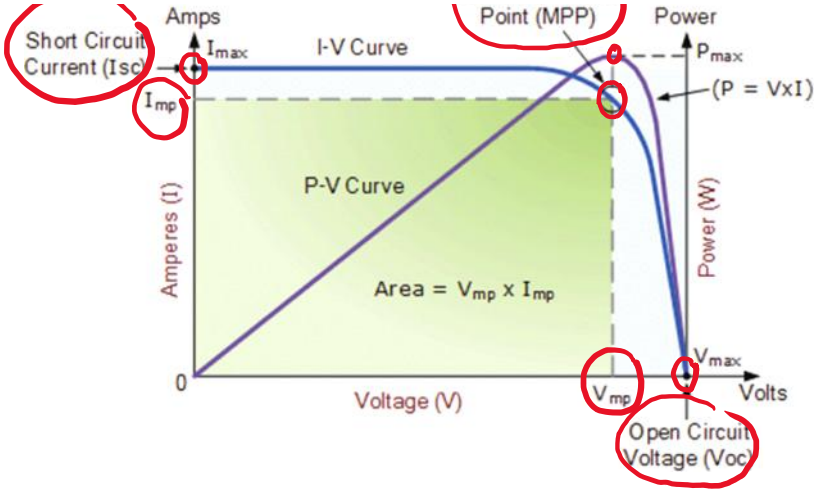
$$FF = \frac{V_{mpp} \times I_{mpp}}{V_{oc} \times I_{sc}}$$

- Efficiency (η):

$$\eta = \frac{V_{mpp} \times I_{mpp}}{G \times A}$$

Area

Irradiance



Example

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Example 1

The current density of a solar cell having an area of 100 cm^2 at STC is given as 35 mA/cm^2. Find out the output current of the solar cell.

(3.5 A)

$I = J A = 35 \times 100 = 3500\ mA = 3.5\ A$

Example 2

A solar PV module is fabricated using 36 solar cells of 155 cm^2 connected in series. The cells have 35 mA/cm^2 current density at STC. Estimate the maximum current produced by the module at STC.

(5.42 A)

$I = J A = 35 \times 155 = 5425\ mA = 5.425\ A$

Example 3

A solar cell gives a current of 0.6 A and voltage of 0.5 V at maximum power point. What is the maximum power point of the solar cell?

(0.3 W)

$P_{mp} = I_{mp} \times V_{mp} = 0.6 \times 0.5 = 0.3\ W$

Example 4

A solar cell having an area of 100 cm^2 gives 3.1 A current at maximum power point and 0.5 V at maximum power point at STC. The cell gives 3.5 A short circuit current and 0.6 V open circuit voltage. What is the maximum power point of the solar cell? Also, find out the efficiency of the cell.

(1.55 W, 15.5%)

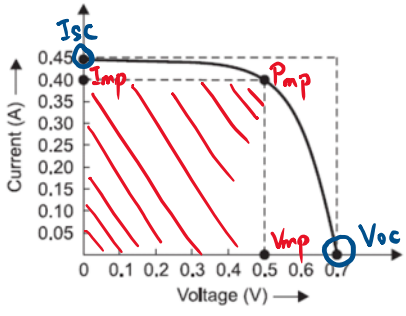
$P_{mp} = I_{mp} \times V_{mp} = 3.1 \times 0.5 = 1.55\ W$

$\eta = \frac{V_{mp} I_{mp}}{G A} = \frac{1.55}{1000 \times 100 \times (10^{-2})^2} \times 100\% = 15.5\%$

Example 5

Refer the I-V curve and find out the Fill Factor for the solar cell.

(63.49%)



$FF = \frac{V_{mp} \times I_{mp}}{V_{oc} \times I_{sc}} \times 100\%$

$= \frac{0.5 \times 0.4}{0.7 \times 0.45} \times 100\%$

$= 63.49\%$

Example 6

Example 6

A solar cell having an area of 25 cm^2 gives a current of 0.85 A and voltage 0.55 V at maximum power point. The short circuit current is 0.9 A and open circuit voltage is 0.65 V. What is the Fill Factor, maximum power point and efficiency of the solar cell? Consider STC.

(0.4675 W, 79.91%, 18.7%)

A → Area; Imp; Vmp; Isc; Voc; G = 1000 W/m^2, T = 25°C

$$P_{mp} = V_{mp} \times I_{mp} = 0.55 \times 0.85 = 0.4675\ W$$

$$FF = \frac{V_{mp} I_{mp}}{V_{oc} I_{sc}} \times 100\% = \frac{0.4675}{0.65 \times 0.9} \times 100\% = 79.91\%$$

$$\eta_{STC} = \frac{V_{mp} I_{mp}}{G A} \times 100\% = \frac{0.4675}{1000 \times 25 \times (10^{-2})^2} \times 100\% = 18.7\%$$

Example 7

Calculate the output power from a module if its efficiency is 22%, input power density is 1000 W/m^2 and area of module is 1.49 m by 0.99 m.

(324.5 W)

Example 8

A solar cell having FF=60% gives 2.5 A current at maximum power point at STC. The cell gives 3 A short circuit current and 0.5 V open circuit voltage. What is the voltage at maximum power point of the solar cell?

(0.36 V)

Example 9

A solar cell having Fill Factor (FF) 68% gives 0.6 V voltage at maximum power point at STC. The cell gives 3 A short circuit current and 0.7 V open circuit voltage. What is the current at maximum power point of the solar cell?

(2.38 A)

Example 10

Calculate the output power from a solar cell if its efficiency is 30, 24, 19, 16 and 12, input power density is 1000 W/m^2, and area of the solar cell is 100 cm^2.

(3W, 2.4W, 1.9W, 1.6W, 1.2W)

Example 11

A Wp rating of a PV module is 230 Wp under STC. What will be the output power of the PV module if the solar radiation intensity is only 400 W/m^2?

(92 Wp)

Example 12

Calculate the output power for solar cells of efficiencies 16%. When the irradiance is say, 1000, 800, 600 and 400 W/m^2 and area of solar cell is 100 cm^2.

(1.6W, 1.28W, 0.96W, 0.64W)

Example 13

A solar cell has maximum power point of 0.3 W. The cell voltage at maximum power point at STC is 0.65 V. What is the current at maximum power point of the solar cell?

(0.46 A)

Factors Affecting Solar Cell Power

There are five common factors that affect the power generated by solar cells. They are:

- The conversion efficiency (η), ✓ Light → Electricity
- The solar irradiation (P_{in}), ✓
- The solar cell area (A) ✓
  - The current output is directly proportional to the cell area.



- The angle at which day light falls (q) ✓
  - Solar cell produces **maximum power** when sunlight falls **perpendicular to the surface of solar cells**. When the light does not perpendicular to solar cells, it always gives less output power.
- The operating temperature (T) ✓
  - When the temperature of solar cell increases; I_{sc} increases while output voltage, efficiency and output power reduces.

PV Technology name	*Typical value of change in parameter value per °C rise in cell temperature from standard test condition (STC) value of 25 °C (+ indicates increase, − indicates decrease)*			
	Temperature coefficient of current (I_{sc})	*Temperature coefficient of voltage (V_{oc})*	*Temperature coefficient of fill factor (FF)*	*Temperature coefficient of power (P_m)*
Crystalline silicon ✓	+0.08%/°C	−0.35%/°C	−0.15%/°C	−0.45%/°C
Cadmium telluride (CdTE)	+0.04%/°C	−0.25%/°C	−0.035%/°C	−0.25%/°C
Double junction amorphous silicon	+0.07%/°C	−0.3%/°C	−0.095%/°C	−0.25%/°C

} TC the smaller the better ✓

https://www.vpsolar.com/download/catalog/Moduli/Panasonic/Panasonic_VBHN300SJ46_datasheet_EN.pdf

- The change in **maximum module voltage** due to increase in cell temperature in PV module will be given as:

$$V_{m(temp)} = V_{m(STC)} - TC_{voltage} \times V_{m(STC)} \times \Delta T$$

$V_{m(temp)}$: $V_{mp(temp)}$; $V_{m(STC)}$: 25°C; $V_{m(STC)}$: 25°C; ΔT: $(T - T_{STC})$

- The change in module's **maximum power output** due to increase in cell temperature in PV module will be given as:

$$P_{max(temp)} = P_{max(STC)} - TC_{power} \times P_{max(STC)} \times \Delta T$$

where TC = temperature coefficient

Example

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Example 1

A manufacturer using advanced crystalline silicon cell technology produces the solar cells with V_{oc} of 0.620 V and V_m of 0.510 V at STC. Cell temperature reaches to about 60°C in operating conditions. The modules are designed to connect in grid connected PV plant. If required operating voltage of module is 13.5V, then estimate the number of cells to be connected in series. Give n per degree centigrade decrease of voltage for crystalline silicon cell = 0.0023 V.

(32)

$TC = -0.0023\ V/°C$

At STC ($T = 25°C$)

No. of modules, $N = \frac{\text{Module voltage}}{\text{Cell voltage}} = \frac{13.5}{0.51} \approx 27$

At $T = 60°C$ (Worst condition)

$V_{mp(60°C)} = V_{mp(25°C)} + \text{change in voltage due to temperature change}$

$= V_{mp(25°C)} + TC_{vmp} \times \Delta T$

$= 0.51 + [-0.0023 \times (60 - 25)]$

$= 0.51 + (-0.0805)$

$= 0.429\ V$

$N = \frac{13.5}{0.429} \approx 32$

Example 2

If the actual operating temperature of the solar cell is 40 °C. The output voltage of a solar cell at standard operating temperature is, say 0.7 V. The output voltage decreases by 2.3 mV/°C. Calculate the new value of output voltage.

(0.67V)

Parameter of crystalline silicon solar cells	Decrease per °C rise in cell temperature from standard test condition (STC) value of 25°C
Voltage	−0.0023 V or −2.3 mV
Power	−0.45%
Efficiency	−0.45%

$V_{mp(40°C)} = V_{mp(25°C)} + \text{change in voltage due to temperature change}$

$= V_{mp(25°C)} + TC_{vmp} \times \Delta T$

$= 0.7 + (-0.0023)(40 - 25)$

$= 0.67\ V$

Example 3

Efficiency of a crystalline silicon solar cell at STC is 15%. What would be its efficiency when cell operating temperature is 60°C?

(12.63%)

Parameter of crystalline silicon solar cells	Decrease per °C rise in cell temperature from standard test condition (STC) value of 25°C
Voltage	−0.0023 V or −2.3 mV
Power	−0.45%
Efficiency	−0.45%

Example 4

The maximum rated power point of a crystalline silicon solar cell is 2.5 Wp. Calculate the maximum power output at 45°C.

(2.275 W)

Parameter of crystalline silicon solar cells	*Decrease per °C rise in cell temperature from standard test condition (STC) value of 25°C*
Voltage	−0.0023 V or −2.3 mV
Power	−0.45%
Efficiency	−0.45%

Example 5

The maximum rated power point of a crystalline silicon solar cell is 75 Wp. Calculate the maximum power output at 55 °C. For crystalline silicon, cell decrease in maximum power per degree centigrade increase in cell temperature is – 0.45%/°C.

(64.87 W)

Example 6

Voltage at maximum power point (Vm) of a crystalline silicon solar PV module at STC is 17 V. What would be the voltage when PV module operating temperature is 60°C? Given temperature coefficient of Voc is -0.35%/°C.

(14.91 V)

Types of Solar Cells

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Common Types of Solar Cells



amorphous silicon

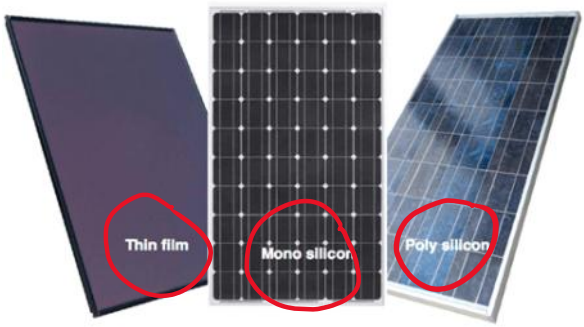


single crystal silicon

(mono)



polycrystalline silicon



Heavy (glass)

Type	Materials	Efficiency (%)	Open-circuit voltage (V)	Short-circuit current density (mA/cm²)	Fill factor (%)
Crystalline Silicon (c-Si) Solar Cell	Mono-crystalline	14 – 16	0.55 – 0.68	30 – 38	70 – 78
	Poly or multicrystalline (mc-Si)	14 – 16	0.55 – 0.68=5	30 – 35	70 – 76
Thin Film Solar Cell	Amorphous Si (a-Si)	6 – 9	0.70 – 1.10	8 – 15	60 – 70
	Cadmium telluride (CdTe)	8 – 11	0.80 – 1.00	15 – 25	60 – 70
	Copper-Indium-Gallium-Selenide (CIGS)	8 – 11	0.50 – 0.70	20 – 30	60 – 70
Multi-Junction Solar Cell	Gallium indium phosphide/ Gallium arsenide/ Germanium (GaInP/GaAs/Ge)	30 – 35	1.00 – 2.50	15 – 35	70 – 85

(Cell)

Light, Curve

(double) (double)

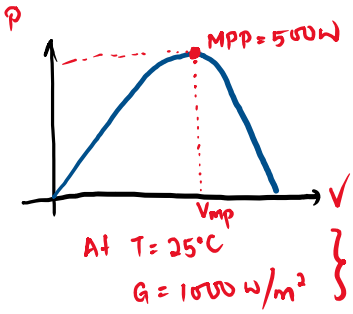
cost ↑

P=↑V I↓

Maximum Power Point Tracking

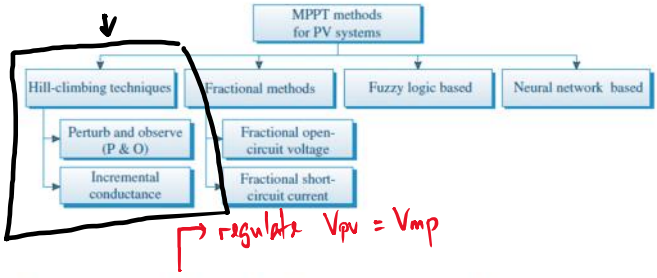
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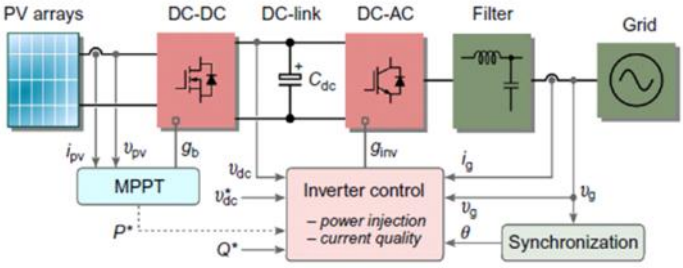
Introduction



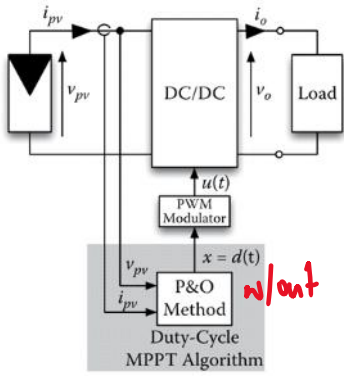
- A PV module has a nonlinear *I–V* characteristic. Power reaches maximum at one point, i.e., the maximum power point (MPP).
- The MPP shifts with irradiance and temperature. (in wind as well)
- MPP tracking (MPPT) is a control technique used in PV converters to continuously adjust operating voltage/current so the PV system always extracts the maximum possible power.
- It works by adjusting the operating point of the PV module via a DC-DC converter. $V_{pv} = V_{mp}$ or $I_{pv} = I_{mp}$
- Perturb and Observe (P&O) and Incremental Conductance (InC) are the most common MPPT algorithms, while advanced AI-based methods are emerging.



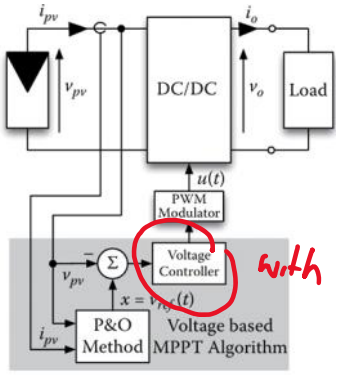




Perturb and Observe

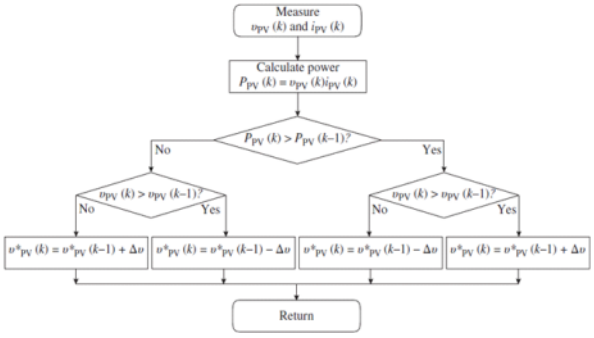


P&O without voltage controller

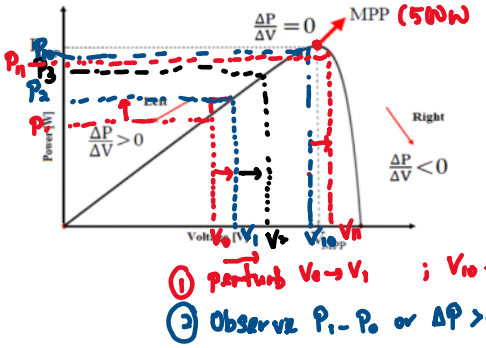


P&O with voltage controller

- The *P-V* curve has a single peak (MPP) under uniform irradiance.
- P&O works by:
  - Perturbing the operating voltage (duty cycle)
  - Observing the change in output power
  - Deciding whether to keep perturbing in the same direction or reverse it
- The perturbation step is selected on the basis of performance trade-off between transient rise time and steady-state conditions.



Disadv → oscillate around MPP

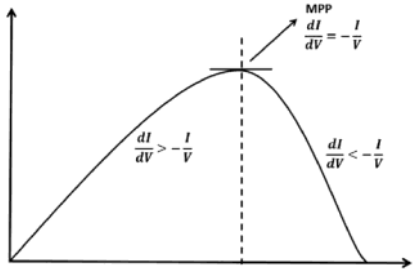


① perturb $V_0 \rightarrow V_1$; $V_{10} \rightarrow V_{11}$, ΔV

② Observe $P_1 - P_0$ or $\Delta P > 0$? ; $\Delta P < 0$

- If $\frac{\Delta P}{\Delta V} = 0$, MPP is reached
- If $\frac{\Delta P}{\Delta V} > 0$, increase voltage
- If $\frac{\Delta P}{\Delta V} < 0$, decrease voltage

Incremental Conductance



Incremental conductance is measured, i.e., $\frac{dI}{dV}$

$$\frac{dP}{dV} = \frac{d(IV)}{dV} = I\frac{dV}{dV} + V\frac{dI}{dV} = I + V\frac{dI}{dV}$$

At MPP,

$$\frac{dP}{dV} = 0$$

$$\frac{dI}{dV} = -\frac{I}{V}$$

$R = \frac{V}{I}$ (resistance)

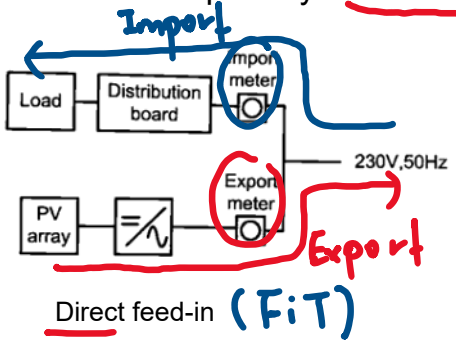
$\sigma = \frac{1}{R} = \frac{I}{V}$

Sizing of PV System

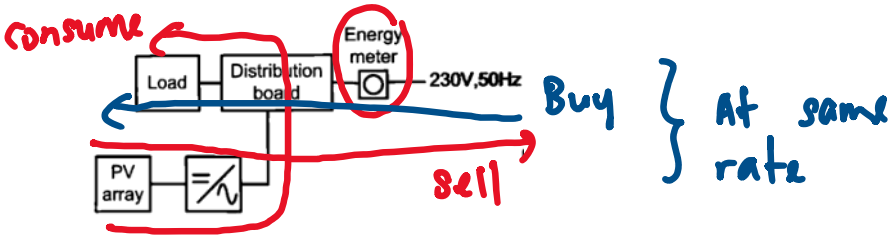
Tuesday, October 6, 2026 10:30 AM

Introduction

- There are two types of connection: Direct feed-in and indirect feed-in.
- In direct feed-in method, power consumption and power generation are segregated and measured independently by import and export meters.
- Indirect feed-in transfers power from PV to the grid via distribution board. This method is adopted by Net Energy Metering (NEM) and Self Consumption schemes.

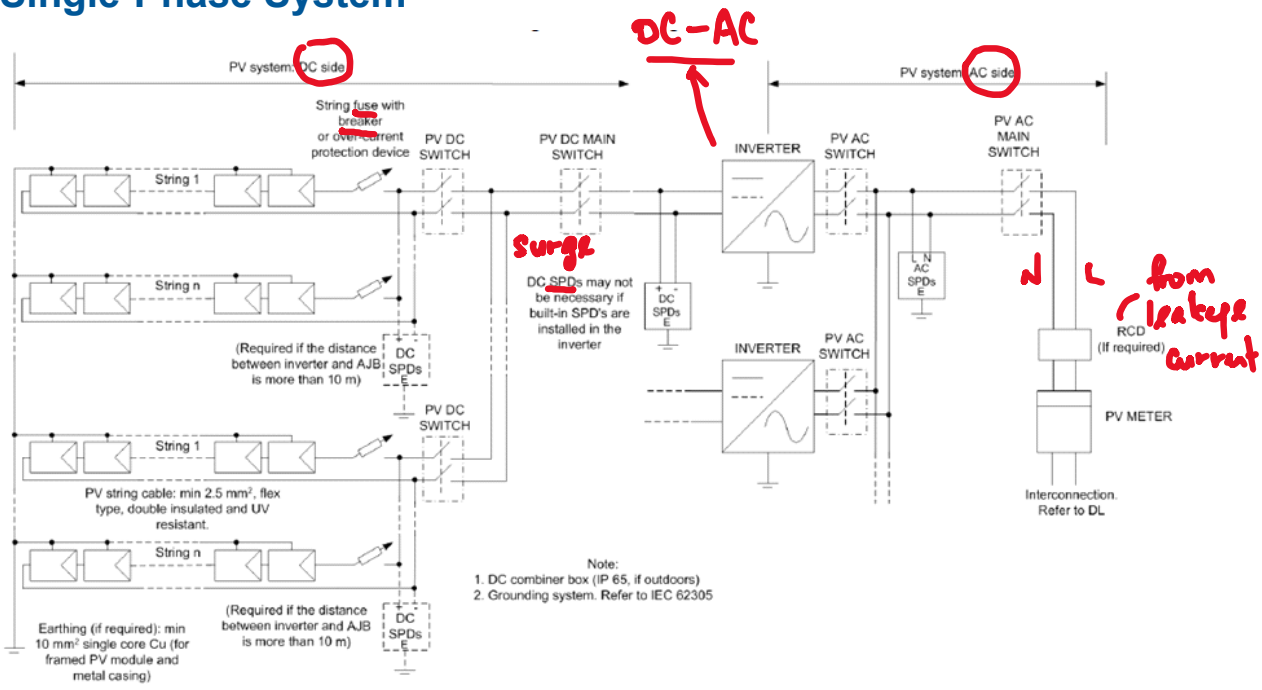


Direct feed-in (FiT)

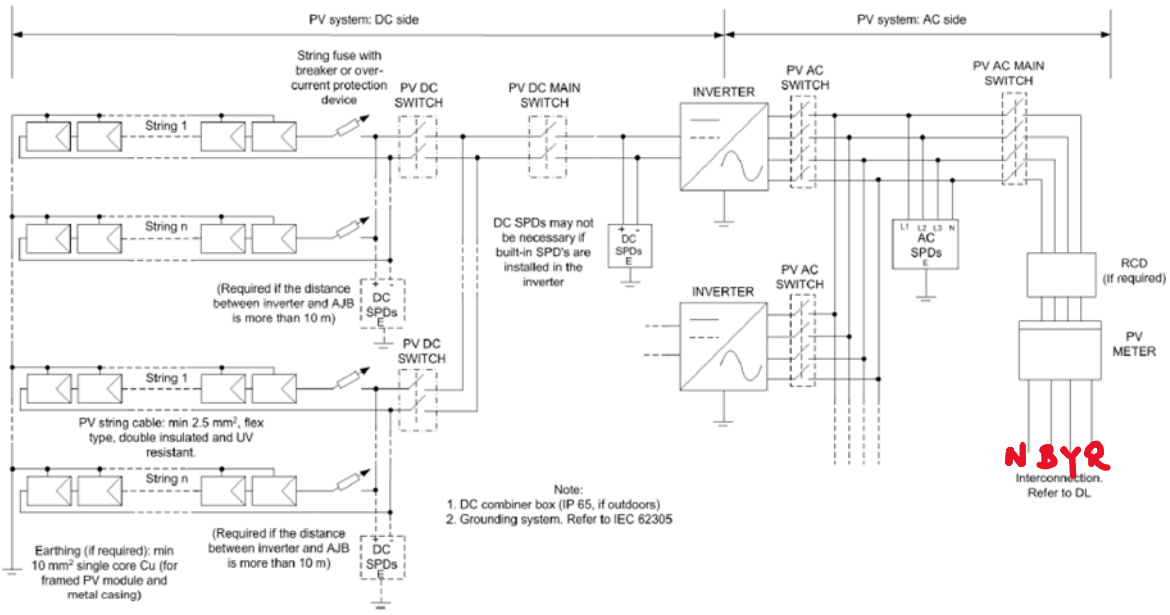


Indirect feed-in

Single-Phase System



Three-Phase System



Sizing of PV System

- The philosophy of design is fundamentally balancing the requirements, constraints, resources and components.
- Three main factors to be considered for rooftop solar photovoltaic (PV) system design are:
  - Energy constraint ✓
  - Space constraint ✓
  - Budget constraint ✓

Design Procedures

- The number of PV modules required are determined based on the energy requirement.
- The following steps that involved are:
  1. Determine the amount of energy to be supplied by the PV.
  2. Determine the minimum power of PV array required.
  3. Determine the minimum number of PV modules required.

PV Array Power

- The energy per annum required to be generated by the PV arrays is

$$E_{req} = \frac{\varepsilon}{100\%} \times \frac{12}{n} \times \sum_{n=1}^{12} E_m$$

$E = Pt$

$P = \frac{E}{t}$

E_{req} = energy required per annum (kWh)
ε = percentage of energy to be supplied using PV (%)
E_m= energy consumption for month m (kWh)
n = number of months with available data

- Based on the annual energy required by the PV array, the minimum power of PV array is calculated as

$$P_{arr_stc} = \frac{E_{req}}{PSH \times n_{inv} \times n_{cab} \times k_{dir} \times k_{mm} \times k_{tem_p_ave}}$$

P_{arr_stc} = maximum power of PV array at STC (Wp)
PSH = peak sun hour per annum (h) → 5 to 6 hours
n_{inv} = maximum efficiency of inverter
n_{cab} = efficiency of DC cable from PV to inverter
k_{dir} = de-rating factor due to dirt
k_{mm} = de-rating factor due to module mis-match
k_{dir} = de-rating factor of power due to average maximum temperature

- The de-rating factor of power due to average maximum temperature can be calculated as:

$$k_{tem_p_ave} = 1 + \left[\left(\frac{\beta_{power}}{100}\right) \times T_{amb}\right]$$

β_{power} is temperature coefficient of power at STC (% per ℃)
T_{amb} is ambient temperatures(℃)

Number of PV Modules

- The minimum number of PV modules is calculated as :

$$N_{\min_mod} = \left[\frac{P_{arr_stc}}{P_{mod_stc}}\right]_{round\ up}$$

$N_{\min_mod}$ is minimum number of PV modules required

P_{arr_stc} is maximum power rating of PV module at STC (Wp)

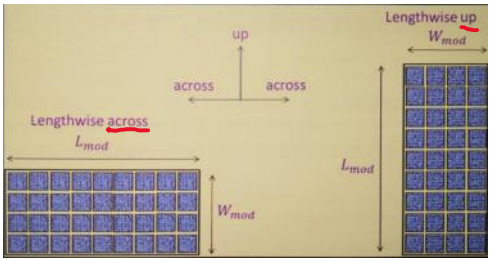
Space Constraint

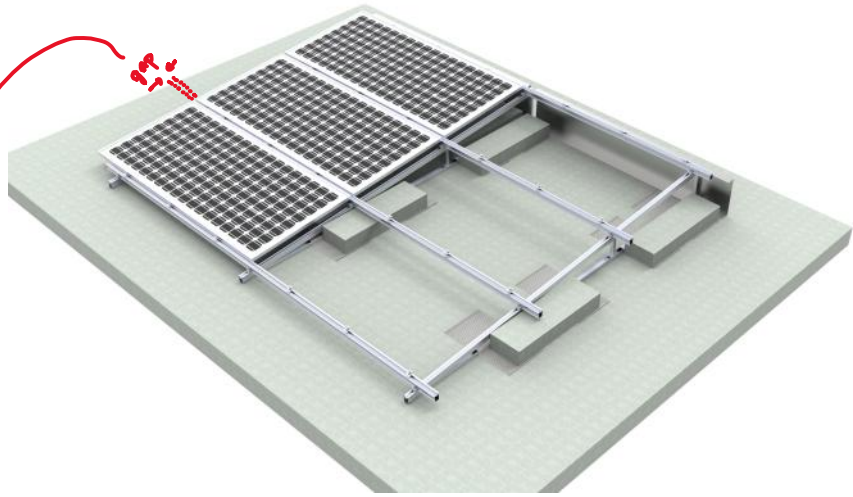
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Introduction

- The second criteria to determine the solar system capacity is space constraint.
- Here, the number of solar modules installed are determined based on th
- There are two possible arrangements of solar modules on the rooftop.
  - Length-wise across
  - Length-wise up

PV Modules Arrangement





Lengthwise Across

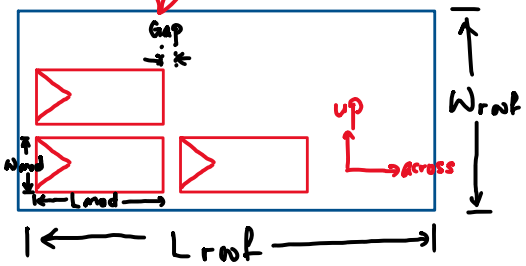
- Under lengthwise across arrangement, the maximum number of solar modules can be installed is

$$N_{\max_mod_across} = N_{across_across} \times N_{across_up}$$

where

$$N_{across_across} = \left[\frac{L_{roof}}{L_{mod} + gap}\right]_{round\ down}$$

$$N_{across_up} = \left[\frac{W_{roof}}{W_{mod} + gap}\right]_{round\ down}$$

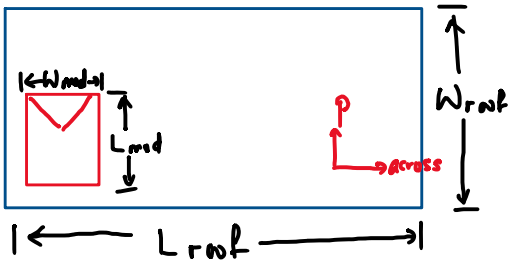


Lengthwise Up

$$N_{up_across} = \left[\frac{L_{roof}}{W_{mod} + gap}\right]_{round\ down}$$

$$N_{up_up} = \left[\frac{W_{roof}}{L_{mod} + gap}\right]_{round\ down}$$

$$N_{\max_mod_up} = N_{up_across} \times N_{up_up}$$



Budget Constraint

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Introduction

- Budget creates a spending plan for money usage and it is crucial to ensure any project can be finished smoothly without any financial problem.
- The cost of the system will be determined by:
  - Total cost PV array power affordable
  - Maximum number of PV modules affordable

& Space constraints

Calculation

- The total PV array power affordable can be calculated as:

$$P_{arr_stc} = \frac{Budget}{K_{ind}}$$

where

Budget is the money allocated for the investment (RM)

K_{ind} is cost index of PV system (RM per Wp)

- The maximum number of PV modules affordable:

$$N_{max_mod} = \left[\frac{P_{array_stc}}{P_{mod_stc}}\right]_{round\ down}$$

where

P_{array_stc} is the maximum power of PV array at STC (Wp)

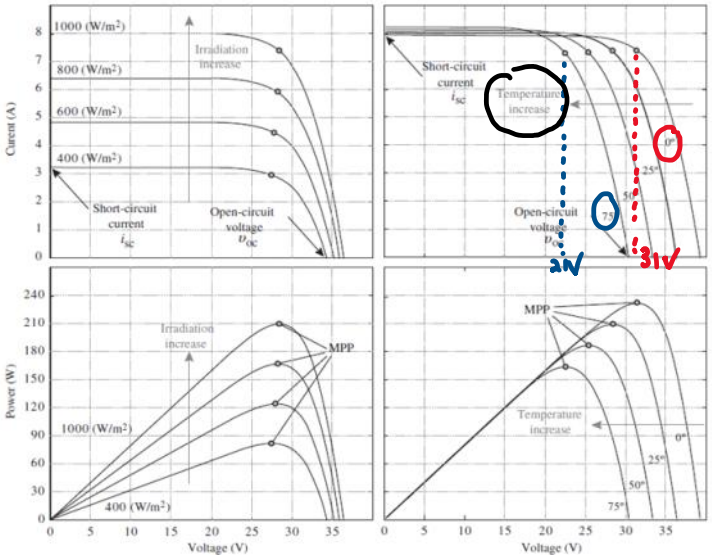
P_{mod_stc} is maximum power of PV module at STC

Sizing of PV Array to Inverter

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Introduction

- The sizing of PV array to inverter is determined by :
  - Range of numbers of modules
  - Extreme limits of voltage
  - String modules → voltage
  - Parallel strings → current
  - Array configuration



Extreme Limits of Voltage

- The maximum voltage produced by PV module:

$$V_{max} = V_{oc_stc} \times \left\{1 + \left[\left(\frac{\beta_{voltage}}{100}\right) \times \Delta T_{\min_stc}\right]\right\}$$

where

V_{max} is the maximum voltage output from PV module (V)

V_{oc_stc} is open circuit voltage at STC (V) → 25°C

$\beta_{voltage}$ is temperature coefficient of voltage (%/°C)

$\Delta T_{\min_stc}$ is difference between minimum operating cell temperature and STC temperature (°C)

- The minimum voltage produced by PV module:

$$V_{min} = V_{mp_stc} \times \left\{1 + \left[\left(\frac{\beta_{voltage}}{100}\right) \times \Delta T_{\max_stc}\right]\right\}$$

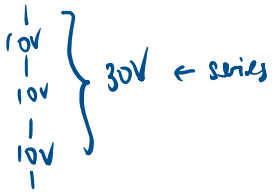
where

V_{min} is the minimum voltage output from PV module (V)

V_{mp_stc} is voltage at maximum power at STC (V)

$\beta_{voltage}$ is temperature coefficient of voltage (%/°C)

ΔT_{max_stc} is difference between maximum operating cell temperature and STC temperature (°C)



String Modules

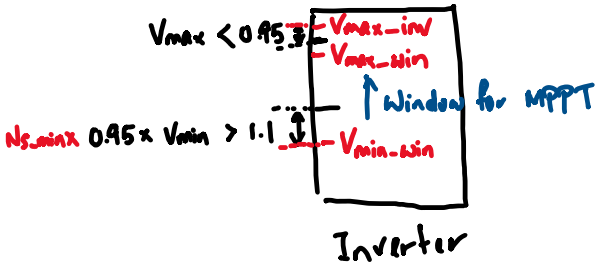
- The range of numbers of PV modules in a string:

$$N_s = [N_{s_min}] .. [N_{s_max}]$$

$$N_{s_min} = \left[\frac{V_{min_inv_win} \times k_3}{V_{min} \times k_4}\right]_{round\ up} = \left[\frac{V_{min_inv_win} \times 1.1}{V_{min} \times 0.95}\right]_{round\ up}$$

$$N_{s_max} = \left[\frac{V_{max_inv} \times k_5}{V_{max}}\right]_{round\ down} = \left[\frac{V_{max_inv} \times 0.95}{V_{max}}\right]_{round\ down}$$

safety



where

(N_s is range of number of PV modules in a string) (series)

N_{s_min} is minimum number of PV modules per string

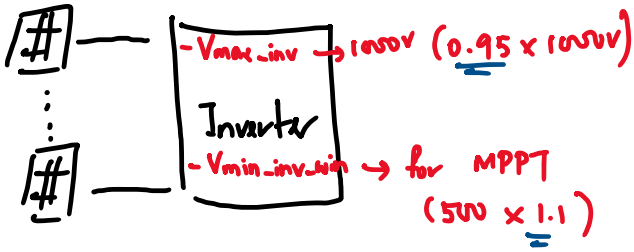
N_{s_max} is maximum number of PV modules per string

$V_{min_inv_win}$ is minimum input voltage window of inverter

V_{max_inv} is maximum input voltage of inverter

k_3 and k_4 are safety margin

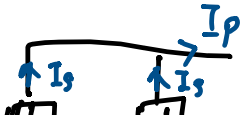
k_5 is cable loss



Parallel Strings

- For poly, mono and HIT panels (Not for thin film):

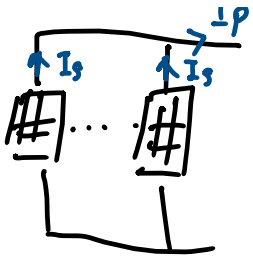
$$N_p = \left[\frac{I_{dc_inv}}{\ldots}\right]$$



- For poly, mono and HIT panels (Not for thin film):

$$N_{p,max} = \left[\frac{I_{dc_inv}}{I_{sc_stc} \times 1.25}\right]_{round\ down}$$

safety



where

N_{p_max} is maximum number of strings in parallel

I_{dc_inv} is maximum input current to the inverter

I_{sc_stc} is short-circuit current of string at STC

- The range of numbers of PV modules in a parallel:

$$N_p = \left[\frac{N_{max}}{N_{s_max}}\right]_{round\ down}$$

Or

$$N_p = \left[\frac{N_{max}}{N_x}\right]_{round\ down}$$

where

$N_{s_min} \leq N_x \leq N_{s_max}$

$1 \leq N_p \leq N_{p_max}$

Array Configuration

- The total array configuration:

$N_t = N_p \times N_s$

3 x 4

parallel

series

where

N_t is total number of PV modules in the array.

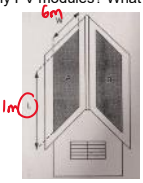
Example

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Example 1

Which type of the following PV module could generate the highest energy based on the given roof space area (L=11m and W=6m)? How many PV modules? What is the total array power? Given gap = 10mm.

- Advent 210
- Aleo S-03 150W
- HIP-210NKHE1 ✓
- BP Solar BP3220N
- Mitsubishi PV-TJ210GA6



(HIP-210NKHE1, 84 modules, 17640 Wp)

Size constraint

Advent 210

Lengthwise across

$$N_{across_across} = \left[\frac{L_{roof}}{L_{mod} + gap}\right]_{round\ down} = \left[\frac{11}{1.68 + 0.01}\right]_{r.d} = 6$$

$$N_{across_up} = \left[\frac{W_{roof}}{W_{mod} + gap}\right]_{round\ down} = \left[\frac{6}{0.99 + 0.01}\right]_{r.d} = 6$$

2 sides of the roof

$N_{max} = 2 \times 6 \times 6$

$= 72$ ✓

$P_{array} = 72 \times 210 = 15.1\ kW$

Lengthwise up

$$N_{up_across} = \left[\frac{L_{roof}}{W_{mod} + gap}\right]_{round\ down} = \left[\frac{11}{0.99 + 0.01}\right]_{r.d} = 11$$

$$N_{up_up} = \left[\frac{W_{roof}}{L_{mod} + gap}\right]_{round\ down} = \left[\frac{6}{1.68 + 0.01}\right]_{r.d} = 3$$

$N_{max} = 2 \times 11 \times 3$

$= 66$

Aleo S-03 150W

Lengthwise Across

$$N_{across_across} = \left[\frac{L_{roof}}{L_{mod} + gap}\right]_{round\ down} = \left[\frac{11}{1.6 + 0.01}\right]_{r.d} = 6\ modules$$

$$N_{across_up} = \left[\frac{W_{roof}}{W_{mod} + gap}\right]_{round\ down} = \left[\frac{6}{0.8 + 0.01}\right]_{r.d} = 7\ modules$$

$N_{max} = 6 \times 7 \times 2$

$= 84$ ✓

$P_{array} = 84 \times 150 = 12.6\ kW$

Lengthwise Up

$$N_{up_across} = \left[\frac{L_{roof}}{W_{mod} + gap}\right]_{round\ down} = \left[\frac{11}{0.8 + 0.01}\right]_{r.d} = 13$$

$$N_{up_up} = \left[\frac{W_{roof}}{L_{mod} + gap}\right]_{round\ down} = \left[\frac{6}{1.6 + 0.01}\right]_{r.d} = 3$$

$N_{max} = 13 \times 3 \times 2$

$= 78$

HIP-210NKHE1 ← Our Choice

Lengthwise Across

$$N_{across_across} = \left[\frac{L_{roof}}{L_{mod} + gap}\right]_{round\ down} = \left[\frac{11}{1.58 + 0.01}\right]_{r.d} = 6\ modules$$

$$N_{across_up} = \left[\frac{W_{roof}}{W_{mod} + gap}\right]_{round\ down} = \left[\frac{6}{0.798 + 0.01}\right]_{r.d} = 7\ modules$$

$N_{max} = 6 \times 7 \times 2 = 84$

$P_{array} = 84 \times 210 = 17.6\ kW$ ↙ Highest power

Lengthwise Up

$$N_{up_across} = \left[\frac{L_{roof}}{W_{mod} + gap}\right]_{round\ down} = \left[\frac{11}{0.798 + 0.01}\right]_{r.d} = 13$$

$$N_{up_up} = \left[\frac{W_{roof}}{L_{mod} + gap}\right]_{round\ down} = \left[\frac{6}{1.58 + 0.01}\right]_{r.d} = 3$$

$N_{max} = 13 \times 3 \times 2 = 78$

Example 2

The maximum budget allocation is RM180,000. How many PV modules is affordable by the client? Given the cost index of PV system is RM10/Wp.

Budget

(85 modules)

Budget Constraint

$$P_{array} = \frac{180{,}000}{10} = 18{,}000\ Wp$$

$$N = \left[\frac{P_{array}}{P_{module}}\right]_{r.d} = \left[\frac{18000}{210}\right]_{r.d} = 85$$

Example 3

L P_{module} Grid L

Example 3

η_{inv} η_{cab}

Assuming maximum efficiency of inverter is 98% and power loss in DC cable is 5%, how many PV modules are required to satisfy the energy requirement if the client requires not more than 90% of his total annual energy consumption to be supplied by PV system?
Given daily PSH is 5 hour, dirt de-rate power is 3% and average temperature is 33°C.
(84 modules)

Monthly energy consumption (kWh)

1	2	3	4	5	6	7	8	9	10	11	12
2400	2300	2200	2300	2500	NA	2300	2400	2300	2200	2250	2300

= 25.15 MWh

1 missing

Energy Constraint

$$E_{req} = \frac{\varepsilon}{100\%} \times \frac{12}{n} \times \sum_{n=1}^{12} E_m = \frac{90}{100} \times \frac{12}{11} \left[2400 + 2300 + \dots 2300\right]$$

n = no. of months where data is available

= 24.987 MWh

$$P_{arr_stc} = \frac{E_{req}}{PSH \times n_{inv} \times n_{cab} \times k_{dir} \times k_{mm} \times k_{tem_p_ave}}$$

NOT given → assume 0.95 (mismatch)

$$= \frac{24.987 \times 10^6}{(5 \times 365) \times 0.98 \times 0.95 \times 0.97 \times 0.95 \times 0.901}$$

= 17.713 kWp

$$k_{tem_p_ave} = 1 + \left[\left(\frac{\beta_{power}}{100}\right) \times T_{amb}\right] = 1 + \left[\frac{-0.3}{100} \times 33\right] = 0.901$$

β_{power}: Temperature coefficient

$$N_{max} = \left[\frac{P_{array}}{P_{mod}}\right]_{r.d.} = \left[\frac{17713}{210}\right]_{r.d.} = 84$$

Space → 84 ✓
Budget → 85
Energy → 84 ✓ ← Choose based on the lowest

$P_{array-stc} = 84 \times 210 = 17640$ Wp
↳ HIP 210 PV module is selected

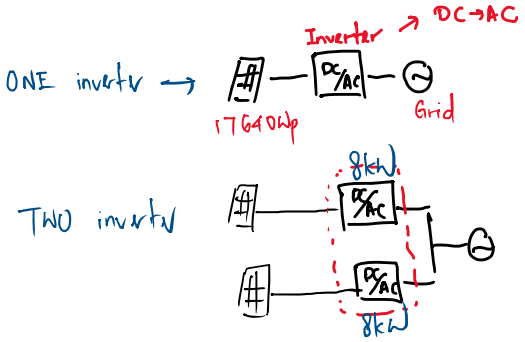
Example 4

$(N_p \times N_s)$

Determine the optimum PV array configuration. Given the following inverters:

- Solarmax 30c
- Fronius IG 15
- Kaco Powador 16.0 TR3
- Solarmax 20c
- Fronius IG Plus 100V-2 ✓

(4 X 10)



A) To select inverter

$0.9\ P_{array-stc} < P_{nom-inv} < 1.0\ P_{array-STC}$

$0.9 \times 17640 < P_{nom-inv} < 1.0 \times 17640$

$15876\ Wp < P_{nom-inv} < 17640\ Wp$

If one inverter is chosen

(i) Solarmax 30c → $P_{out} = 30$ kW
↳ It is too large

(ii) Fronius IG 15 → $P_{out} = 1.5$kW
↳ It is too small

If two inverters are chosen

$7938\ Wp < P_{nom-inv} < 8820\ Wp$

(i) Solarmax 30c → $P_{out} = 30$kW
↳ It is too large

(ii) Fronius IG 15 → Pout = 1.5kW
↳ It is too small

(iii) Kaco Powador 16.0 TRS → Pout = 13.5 kW
↳ It is too small

(iv) Fronius IG Plus 100 V-2 → Pout = 8kW
↳ It is too small

(i) Solarmax 50C → Pout = 30 kW
↳ It is too large

(ii) Fronius IG Plus 100 V-2 → Pout = 8kW
↳ It is suitable ✓ Our choice #

B) PV configuration

(i) $N_{min} = \left[\frac{P_{inv\text{-}nom}}{1.0 \times P_{mod\text{-}STC}}\right]_{r.d.} = \left[\frac{8000}{1.0 \times 210}\right]_{r.u.} = 39$ modules ✓

$N_{max} = \left[\frac{P_{inv\text{-}nom}}{0.9 \times P_{mod\text{-}STC}}\right]_{r.d.} = \left[\frac{8000}{0.9 \times 210}\right]_{r.d.} = 42$ modules ✓

$N_T \approx 39 < N_T \leq 42$

(ii) $V_{max} = V_{oc_stc} \times \left\{1 + \left[\left(\frac{\beta_{voltage}}{100}\right) \times \Delta T_{min_stc}\right]\right\}$ (β_{voc} (Voc))

$\beta_{voc} = -0.127$ V/°C $= \frac{-0.127}{50.9} \times 100\% = -0.2495$ %/°C ($V_{oc,STC}$)

$= 50.9 \times \left\{1 + \left[\frac{-0.2495}{100} \times (20 - 25)\right]\right\}$ (T_{min})

$= 51.535$ V

$V_{min} = V_{mp_stc} \times \left\{1 + \left[\left(\frac{\beta_{voltage}}{100}\right) \times \Delta T_{max_stc}\right]\right\}$ (P_{vmp}, T_{max})

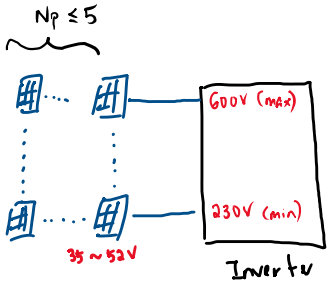
$= 41.3 \times \left\{1 + \left[\frac{-0.3}{100} \times (75 - 25)\right]\right\}$

$= 35.105$ V

(iii) $N_{s_min} = \left[\frac{V_{min_inv_win} \times 1.1}{V_{min} \times 0.95}\right]_{round\ up} = \left[\frac{230 \times 1.1}{35.105 \times 0.95}\right]_{r.u} = 8$ modules (inverter)

$N_{s_max} = \left[\frac{V_{max_inv} \times 0.95}{V_{max}}\right]_{round\ down} = \left[\frac{600 \times 0.95}{51.535}\right]_{r.d} = 11$ modules (inverter)

$8 < N_S \leq 11$ modules



(iv) $N_{p,max} = \left[\frac{I_{dc_inv}}{I_{sc_stc} \times 1.25}\right]_{round\ down} = \left[\frac{37.1}{5.57 \times 1.25}\right]_{r.d} = 5$ strings in parallel

(inverter; 1.25 = safety factor)

$N_P \mid N_S \mid N_T$ ✓

$N_T \approx 39 < N_T \leq 42$ ✓

$V_{min} = 35.105\ V$

$V_{max} = 51.535\ V$

$8 < N_S \leq 11$ modules ✓

$N_{p,max} = 5$ strings ✓

N_p	N_s	N_T	
5	11	55	
5	10	50	
5	9	45	X out of range
4	11	44	
4	10	40	✓

PV configuration is 4 (parallel) × 10 (series)

Introduction

- The common safety standards of PV module products include IEC-61215, IEC-61646, IEC-61730, and UL-1703.
- As well as the PV cells, other components used for PV module construction should be evaluated to ensure their safety, reliability, quality, and performance.
- These include polymeric materials, junction boxes, connectors, and connecting cables.

Standards

Standards	Title
IEC 61215	Crystalline silicon terrestrial photovoltaic (PV) modules – Design qualification and type approval
IEC 61646	Thin-film terrestrial photovoltaic (PV) modules– Design qualification and type approval
IEC 61730	Photovoltaic (PV) module safety qualification: Part 1– Requirements for construction Part 2– Requirements for testing
UL 1703	Standard for flat-plate photovoltaic modules and panels
UL 2703	Standard for mounting systems, mounting devices, clamping/ retention devices, and ground lugs for use with flat-plate photovoltaic modules and panels
UL 3703	Standard for solar trackers for use with distributed energy resources

Regulation Bodies

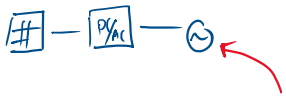
- The International Electrotechnical Commission (IEC) is a not-for-profit, non-governmental organization, which publishes consensus-based standards and manages conformity assessment systems for electrical and electronic products.
- In Europe, the leading organizations for certification are the Germany-based TÜV Rheinland and VDE, the Association for Electrical, Electronic, and Information Technologies.
- In North America, the UL-1703 standard is commonly applied and PV modules are validated by either the Underwriters Laboratories (UL) or the Electrical Testing Laboratory (ETL), a division of the Intertek Group.

Certification

- For certification, PV module manufacturers must submit the product specification, full material lists, product samples, and related documents to the certification agencies.
- After an initial evaluation, the certification body starts lab tests to assess construction and performance.
- This comprehensive evaluation guarantees the product's performance, quality, and safety. When all tests have been passed, the certification can be issued.
- The end users can search for certified products through the online databases provided by the certification agencies.

Interconnection Standards

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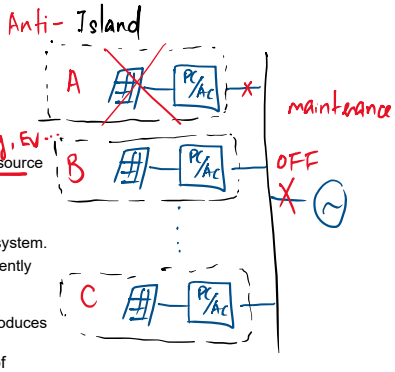
Introduction

- PV converters must meet safety standards of UL 1741 and IEC 62109 for grid interconnection.
- UL 1741 was first introduced in the late 1990s and revised in 2001. It is the product safety standard that defines the testing and certification requirements for inverters, converters, controllers, and other interconnection system equipment.
- In most jurisdictions of the USA and Canada, compliance with UL 1741 is mandatory for any PV power system that will be connected to the utility grid.
- The performance testing of UL 1741 is comprehensive, which will not be covered here.

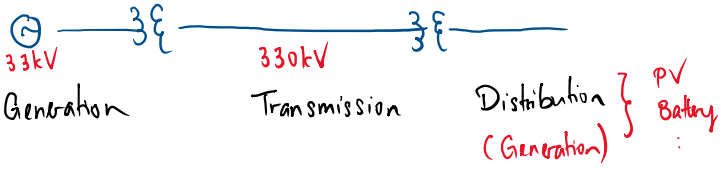
Standards

Standards	Title
UL1741	Standard for Inverters, Converters, Controllers and Interconnection System Equipment for Use With Distributed Energy Resources
IEC 62109-1:2010	Safety of power converters for use in photovoltaic power systems–Part 1: General requirements
IEC 62109-2:2011	Safety of power converters for use in photovoltaic power systems–Part 2: Particular requirements for inverters

Anti-Islanding



- In grid-connected PV systems, the term of islanding refers to the condition in which the distributed resource (DR) continues to supply electric power even though the electric utility is malfunctioning.
- Islanding involves a potential of safety risk to utility workers.
- Therefore, the function of anti-islanding or islanding protection is mandatory in UL 1741 against the continuous power supply from the inverter, which is isolated from the remainder of the electric utility system.
- To meet the requirement, research has been conducted to detect the grid failure accurately and efficiently regardless of the presence of harmonics and noise.
- Anti-islanding includes both passive and active methods. The active anti-islanding algorithm is more effective than passive counterparts to avoid a non-detection zone, but the conventional algorithm introduces disturbance to the utility grid.
- The PV inverters need to disconnect from the grid in case of abnormal grid conditions in terms of voltage and frequency.
- This response is to ensure the safety of utility maintenance personnel and the general public, as well as to avoid damage to connected equipment, including the PV system.
- This requirement is commonly referred to as “anti-islanding protection.”



Voltage Deviations

(400V) (415V)
230V or 240V

- All discussions regarding system voltage refer to the local nominal voltage.
- The voltages in RMS are measured at the point of utility connection.
- The disconnection time refers to the time between the abnormal condition occurring and the inverter ceasing to energize the utility line.
- The inverter controls shall actually remain connected to the utility to allow sensing of utility electrical conditions for use by the ‘reconnect’ feature.
- The purpose of the allowed time delay is to ride through short-term disturbances to avoid excessive nuisance tripping.

Temporary fault
three branches

IEEE 1547		IEC 61727		VDE0126-1-1	
Voltage range (%)	Disconnection time (s)	Voltage range (%)	Disconnection time (s)	Voltage range (%)	Disconnection time (s)
$V < 50$	0.16 ✓	$V < 50$	0.10 ✓	$110 \le V < 85$	0.2 ✓
$50 \le V < 88$	2.00 ✓	$50 \le V < 85$	2.00 ✓		
$110 \le V < 120$	1.00	$110 \le V < 135$	2.00 ✓		
$V \ge 120$	0.16	$V \ge 135$	0.05 ✓		

Frequency Deviations

(50Hz)

- The purpose of the allowed range and time delay is to ride through short-term disturbances to avoid excessive nuisance tripping in weak-grid situations.

IEEE 1547		IEC 61727		VDE0126-1-1	
Frequency (Hz)	Disconnection time (s)	Frequency (Hz)	Disconnection time (s)	Frequency (Hz)	Disconnection time (s)
$59.3 < f < 60.5$*	0.16	$f_n - 1 < f < f_n + 1$	0.2	$47.5 \le f < 50.2$	0.2

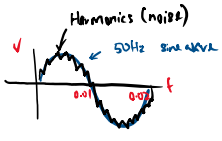
*For IEEE 1547-2018, 57 Hz < f < 61.8 Hz

Note: While IEC 61727 sets these general guidelines, specific local grid codes (such as EN50438, VDE, or local TNB/MEA regulations) may adopt slightly tighter or different tolerance values (e.g., 50Hz ±2% or ±0.5 Hz) for the final interconnection agreement.

Reconnection After Trip

- After a disconnection caused by abnormal voltage or frequency conditions the inverter can be reconnected only in the conditions.

IEEE 1547	IEC 61727	VDE0126-1-1
$88 < V < 110$ (%) **AND** $59.3 < f < 60.5$ (Hz)	$88 < V < 110$ (%) ✓ **AND** $f_n - 1 < f < f_n + 1$ (Hz) ✓ **AND** Minimum delay of 3 minutes ✓	



Power Quality

How clean is your signal?
How high is the harmonics?

- The quality of the power provided by the photovoltaic system for the local AC loads and for the power delivered to the utility is governed by practices and standards on voltage, flicker, frequency, harmonics and power factor.
- Deviation from these standards represents out-of-bounds conditions and may require disconnection of the photovoltaic system from the utility.

No DC current → saturate transformer

DC Current Injection

- DC current injection in the utility can saturate the distribution transformers, leading to over heating and trips.
- For the conventional PV systems with galvanic isolation, this problem is minimized, but with the new generation of transformerless PV inverters increased attention is required in this matter.
- For IEEE 1574 and IEC 61727 the DC component of the current should be measured by using harmonic analysis (FFT) and there is no maximum trip time condition.
- During the test the measured DC component should be below the limits for different loading conditions (1/3, 2/3 and 3/3 of the nominal load).
- For VDE 0126-1-1 this condition requires a special designed current sensor that can detect this threshold and disconnect within the required trip time.

DC current injection limitation

IEEE 1547	IEC 61727	VDE0126-1-1
I_{DC} < 0.5 % I_{rated}	I_{DC} < 1 % I_{rated}	I_{DC} < 1 A Maximum trip time 0.2 s

Current Harmonics

- The PV system output should have low current-distortion levels to ensure that no adverse effects are caused to other equipment connected to the utility system.

Maximum current harmonics in odd order

IEEE 1547 and IEC 61727						
Individual harmonic order (odd) (%)	h <11	11≤ h <17	17≤ h < 23	23≤ h <35	h ≤ 35	Total harmonic distortion (THD) (%)
	4.0	2.0	1.5	0.6	0.3	5.0

Intermittent Issues

Monday, October 12, 2026 8:35 AM

Introduction

- Renewable energy sources provide clean, sustainable electricity but suffer from intermittency and variability due to weather and diurnal cycles.
- These intermittency and variability create several issues to the grid:
  - **Intermittency**: Power output depends on solar irradiance.
  - **Mismatch between demand and generation**: PV often generates maximum power at midday, while demand peaks in the evening.
  - **Grid stability issues**: Sudden fluctuations affect frequency, voltage, and power quality.

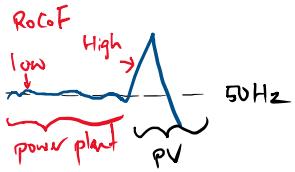
Power Imbalance

- In an AC power system, power balance is met when
  $P_{gen}(t) = P_{load}(t) + P_{loss}(t)$
  $P_{gen}(t)$ = total real power generated
  $P_{load}(t)$ = total real power consumed
  $P_{loss}(t)$ = losses in transmission/distribution
- There is power imbalance if
  $\Delta P(t) = P_{gen}(t) - P_{load}(t) - P_{loss}(t) \neq 0$
- This imbalance ΔP causes acceleration or deceleration of synchronous machine rotors (or equivalently a mismatch in inverter control setpoints).
- Given the swing equation

$$2H\frac{d\omega}{dt} = P_m - P_e - D\Delta\omega$$

$$RoCoF = \frac{df}{dt} \approx \frac{\Delta P}{2H_{sys}S_{base}}$$

where
$RoCoF$ = rate of change of frequency
H_{sys} = system inertia constant (s),
ω = rotor electrical angular speed (rad/s),
P_m = mechanical input power,
P_e = electrical output power,
D = damping factor.
S_{base} = apparent power (MVA)



50Hz ± 1Hz

- If $P_{gen} > P_{load}$: extra energy accelerates rotors → frequency rises.
- If $P_{gen} < P_{load}$: rotors slow down → frequency falls.
- The large inertia constant H_{sys} of synchronous machines resists fast changes.
- With high PV penetration (low inertia), H_{sys} is small, so for the same imbalance ΔP, the $RoCoF$ is much higher.
- In short, the grid with high penetration of renewable energy is sensitive to frequency deviation.

Example

- System base = 1000 MW,
- Inertia constant H_{sys} = 3 s (low-inertia system),
- Sudden imbalance: ΔP = −100 MW

$$\frac{df}{dt} \approx \frac{\Delta P}{2HS_{base}} = \frac{-100}{2\times3\times1000} = -0.01607\ Hz/s$$

- So, frequency drops at ~0.017 Hz per second.
- In a traditional system with H_{sys} = 6, $RoCoF$ would be only ~0.008 Hz/s. which is much slower.

Ramp Rate Control



- PV power P_{PV} output fluctuates rapidly, creating "ramps" or quick changes in power.
- These rapid changes in generation (ramping) burdens the grid and forces conventional generators to ramp up and down quickly to meet the changing demand.
- The net power P_{net} is
  $P_{net}(t) = P_{load}(t) - P_{PV}(t)$
- Power ramp rate required from dispatchable units is:

$$R(t) = \frac{dP_{net}}{dt} = \frac{dP_{load}}{dt} - \frac{dP_{PV}}{dt}$$

power plant cannot response immediately

- If $\frac{dP_{PV}}{dt}$ is large negative (PV drops quickly), dispatchable units must provide $-\frac{dP_{PV}}{dt}$ MW/min to compensate the drop.
- These power fluctuations lead to grid instability, voltage fluctuations, and frequency deviations, which can harm sensitive equipment.

Example

- Assume large PV ramp down late afternoon, $\frac{dP_{PV}}{dt}$ = -50 MW/10 min.
- Then dispatchable units must ramp +5 MW/min.
- However, a thermal plant has typical maximum ramp rate of 2 MW/min.
- There is a mismatch, and a need for storage.

Power, voltage, frequency deviation

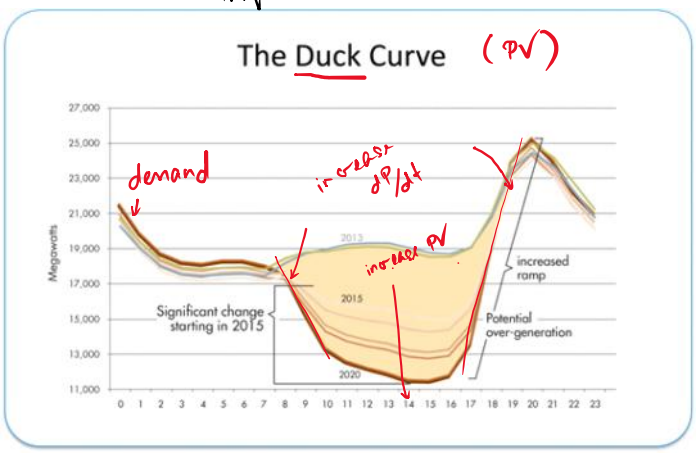


Image courtesy of Green Tech Media

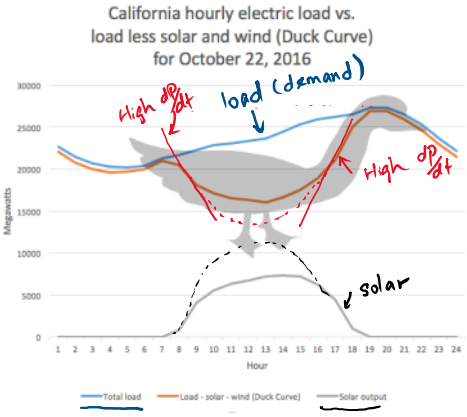


Image courtesy of Wikipedia

Because **power plants can't respond quickly enough** to manage the **sharp evening increase in usage**, expensive and dirty power sources are kept idling during the day to step-in when the demand is on. Therefore, depending on your energy retailer, the cost of using electricity during peak times is growing.

Voltage Rise and Reverse Power Flow

- When distributed PV exports power back toward the substation, voltage at a customer point can increase.
- Using a simple single-line model, the approximate complex voltage drop ΔV along a line is:

$$\Delta V \approx \frac{RP + XQ}{V}$$

where R, X are line resistance and reactance, P, Q are real and reactive power flowing (positive when exported toward the substation), and V is nominal voltage.

- If PV exports real power $P > 0$ back into the feeder, the sign of ΔV can be positive, i.e., voltage rise.

Example

Compute the voltage rise for the following power network:

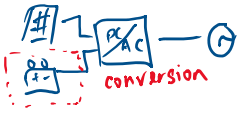
- R = 0.2 Ω, X = 0.8 Ω (short radial feeder numbers)
- PV injects P = 50 kW = 0.05 MW, Q = 10 kVAr = 0.01 MVAr
- Nominal voltage V_s = 400 V (line-to-line for low voltage network)

$$\Delta V \approx \frac{RP + XQ}{V} = \frac{(0.2\times 50000)\times(0.8\times 10000)}{400} = 45\ V$$

So the local voltage magnitude would increase by about 45 V (≈ 11.25% of 400 V).

Energy Storage

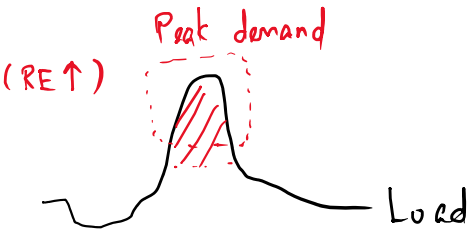
Monday, October 12, 2026 8:40 AM



Role of Energy Storage System

To ensure reliability, stability, and quality of utility grid, energy storage systems (ESS) are essential.

- **Energy Balancing:** Stores excess energy when generation > demand, releases it when demand > generation.
- **Grid Stability:** Provides frequency and voltage regulation to reduce fluctuations.
- **Peak Shaving & Load Shifting:** Store excess PV energy during the day and release during peak demand.
- **Backup Power:** Ensures continuous supply during outages or low generation.
- **Maximizing Renewable Penetration:** Enables higher use of renewables without compromising reliability.



TOU → Non peak → 10pm → 10am (cheap)

peak → 10am → 10pm (expensive)

Performance Metrics

- Energy storage system has following technical requirement:
  - Round-trip efficiency
  - Response time
  - Lifetime and cycling
  - Sizing

years; charge & discharge

Round-Trip Efficiency

I²R

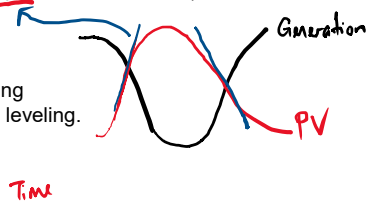
- Losses occur due to internal resistance, conversion devices, and heat dissipation.
- Round-trip efficiency is the **ratio of the energy retrieved from the storage system during discharge to the energy put into it during charging**.

$$\eta = \frac{E_{discharge}}{E_{charge}} \times 100\%$$

- Higher efficiency means **less energy wasted**.
- Among energy storage options, compressed-air energy storage (CAES) has the lowest reported efficiency, and Li-ion batteries have the highest.
- For energy storage coupled with photovoltaics, efficiencies of less than 75% are unlikely to be cost-effective.

Response Time

- Response time is the time taken for the storage system to **respond and deliver power when required**.
- Some applications (like grid frequency regulation) demand very fast response, while others (like seasonal storage) can tolerate slower response.
- Examples
  - Supercapacitors: **Very fast**, good for frequency regulation
  - Lithium-ion batteries: **Fast**, suitable for load shifting and grid balancing
  - Pumped hydro storage: **Slower start-up**, suited for large-scale load leveling.



Lifetime and Cycling

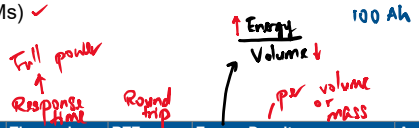
- The expected **operational life of the ESS**, usually measured **in years or number of charge-discharge cycles** before performance degrades below a usable level (often 70–80% of original capacity).
  - Calendar life: Ageing over time even without cycling.
  - Cycle life: How many times the ESS can be charged and discharged.
- Examples
  - Lithium-ion batteries: ~3,000–10,000 cycles (10–15 years).
  - Lead-acid batteries: ~500–1,500 cycles (3–5 years).
  - Flow batteries (e.g., Vanadium Redox): ~10,000+ cycles (15–20 years).
  - Pumped hydro: 30–60 years lifetime.

Sizing

- Sizing is the process of determining the **required energy capacity (kWh or MWh) and power rating (kW or MW)** of the ESS to meet the intended application.
- **Power rating (kW/MW)** determines the instantaneous power the system can deliver.
- **Energy capacity (kWh/MWh)** dictates the duration it can sustain that power (duration of storage).

Types of Energy Storage System

- **Mechanical Storage**
  - Pumped Hydro Energy Storage (PHES)
  - Compressed Air Energy Storage (CAES)
  - Flywheels
- **Electrochemical Storage (Batteries)**
  - Lead-Acid Batteries
  - Lithium-Ion Batteries
  - Flow Batteries (Vanadium Redox, Zinc-Bromine)
  - Sodium-Sulfur (NaS) Batteries
- **Electrical Storage**
  - Supercapacitors
  - Superconducting Magnetic Energy Storage (SMES)
- **Thermal Storage**
  - Molten Salt
  - Phase Change Materials (PCMs)



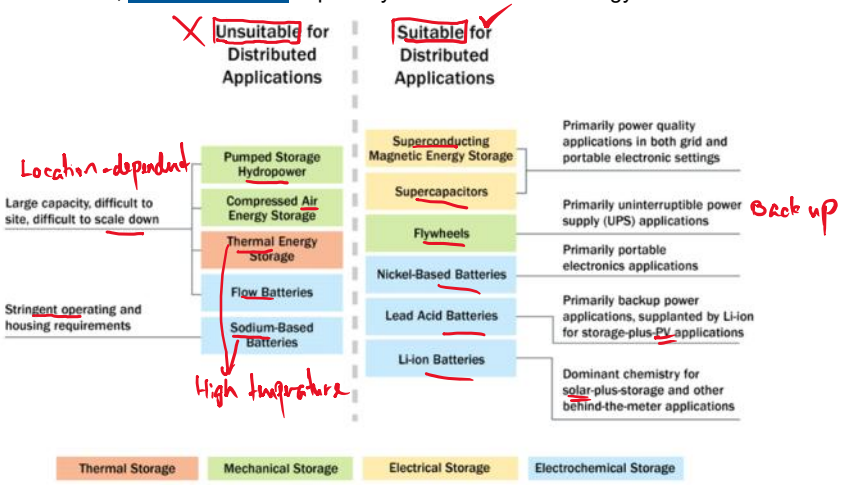
Storage Type	Examples	Timescale	RTE Efficiency (%)	Energy Density	Applications
Electrochemical	Lithium-ion, Lead-acid, Flow batteries	Short (min–days)	70–95	High (Li-ion), Low (Lead-acid)	Scalable, good for PV
Mechanical	PHES, CAES, Flywheels	Medium (hours–days)	60–85	Very high (PHS), moderate (Flywheel)	Grid-scale, large footprint
Thermal	Molten salts, Phase change materials	Medium (hours–days)	50–90	Moderate	Better for solar thermal, not PV
Chemical	Hydrogen, Synthetic fuels	Long (days–months)	30–50	Very high (as fuel)	Expensive, Good for seasonal storage
Electrical	Supercapacitors	Very short (sec–min)	90–98	Very low	Best for fast response, not bulk storage

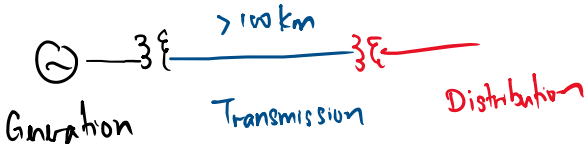
Battery Energy Storage System

Monday, October 12, 2026 8:43 AM

Why BESS?

- **Battery storage** is considered the most suitable form of energy storage for PV systems compared to other ESS because it offers **high efficiency, scalability, and flexibility** in deployment.
- Unlike mechanical ESS (pumped hydro or compressed air), which are **location-dependent** and suited mainly for **large-scale applications**, batteries can be installed directly **alongside PV arrays** in residential, commercial, or utility-scale systems.
- They operate on DC, which matches the output of PV panels, avoiding additional conversion losses and simplifying integration.
- Batteries also respond almost **instantaneously** to fluctuations in solar generation, making them highly effective for smoothing power output during rapid changes in irradiance.
- Furthermore, **declining costs** especially in lithium-ion technology have made BESS increasingly affordable and widely adopted.





BESS Control



Battery Controller

- Prevent under/overcharging
- Balance multiple stacks
- Regulate state of charge (SoC) and monitor temperature



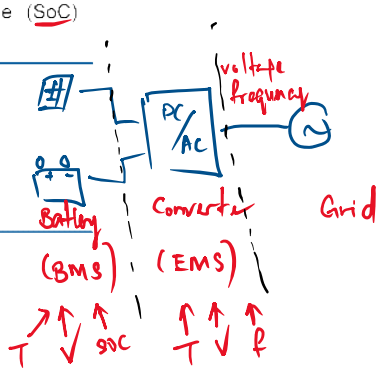
Energy Storage System Controller

- Transient stability
  - Voltage control
  - Frequency control
- Island operation
- Renewable smoothing

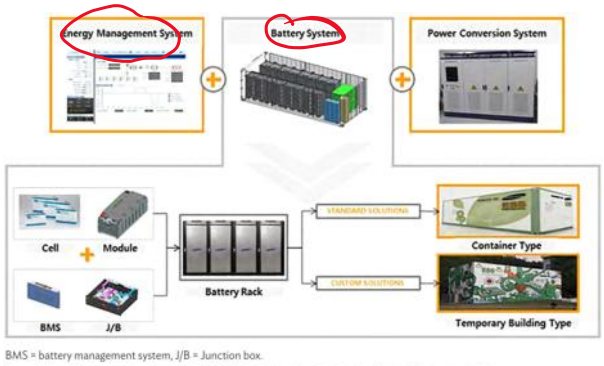


Grid Controller

- Demand response
- Peak shaving
- Time shifting
- Ancillary services.



BESS Block Diagram



BMS = battery management system, J/B = Junction box.

Source: Korea Battery Industry Association 2017 "Energy storage system technology and business model".

- **The battery system** consists of the **battery pack**, which connects **multiple cells** to appropriate voltage and capacity.
- **The battery management (system BMS)** **protects the cells** from harmful operation, in terms of voltage, temperature, and current, to achieve reliable and safe operation, and balances varying cell states-of-charge (SOCs) within a serial connection.
- **The battery thermal management system (B-TMS)** controls the **temperature of the cells** according to their specifications in terms of absolute values and temperature gradients within the pack.

- The components required for the reliable operation of the overall system are system control and monitoring, the energy management system (EMS), and system thermal management.
- System control and monitoring is general **(IT) monitoring**, which is partly combined into the overall supervisory control and data acquisition (SCADA) system but may also include fire and distribution.
- System thermal management controls all functions related to the heating, ventilation, and air-conditioning of the containment system.
- The power electronics can be grouped into the conversion unit, which **converts the power flow between the grid and the battery**, and the required control and monitoring components—voltage sensing units and thermal management of power electronics components (fan cooling).

Types of BESS

charge and discharge
(**round trip**)

	Energy density (kW/kg)	Round Trip Efficiency (%)	Life Span (years)	Eco-friendliness
Li-ion	1st (150–250)	1st 95	1st (10–15)	Eco-friendliness
NaS	2nd (125–150)	2nd (75–85)	3rd (10–15)	X
Flow	2nd (60–80)	2nd (70–75)	3rd (20–25)	X
Ni-Cd	4th (40–60)	4th (60–80)	4th (5–10)	X
Lead Acid	5th (30–50)	5th (60–70)	5th (3–6)	X

Li-ion = lithium-ion, Na-S = sodium-sulfur, Ni-Cd = nickel-cadmium.
Source: Korea Battery Industry Association 2017 "Energy storage system technology and business model".

NaS, Flow, Ni-Cd, and Lead-Acid batteries are less eco-friendly than Li-ion batteries primarily due to the high toxicity of their raw materials, hazardous waste potential, and lower energy efficiency, which leads to **higher carbon emissions** over their lifecycles

Vented Lead–Acid Battery



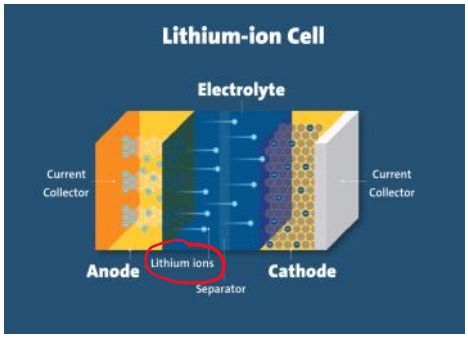
- Oldest battery type, in commercial use for over 100 years.
- Operate from the action of lead and lead oxide electrodes, typically with sulphuric acid electrolyte.
- **Advantages**
  - Can deliver high currents.
  - Easily recycled through existing channels.
- **Disadvantages**
  - Uses lead, which is harmful to the environment in disposal and potentially harmful to health for maintainers.
  - Not suited to complete discharge, which may irreversibly damage the battery.
  - Charges slowly.
  - Releases hydrogen gas when charging, so fresh air ventilation is required. Electrolyte maintenance is required. The acid electrolyte is caustic.

Valve-Regulated Lead–Acid Battery



- Sometimes known as 'sealed' batteries because they require no maintenance.
- Principally, there are two varieties: 'acid-gel' electrolyte or absorbent glass mat. Both operate on a similar electrochemical principal to VLA.
- **Advantages**
  - No maintenance required.
  - Can deliver high currents.
  - 'Deep cycle' versions are available that permit more complete discharge than VLA.
  - Less susceptible to electrolyte leakage if broken.
  - Easily recycled through existing channels.
- **Disadvantages**
  - Uses lead, which is harmful to the environment in disposal.
  - Overcharging can damage the battery.
  - In overcharge or fault conditions, certain VRLA batteries may produce hydrogen, and fresh air ventilation is therefore required.

Lithium-Ion (Li-Ion) Battery



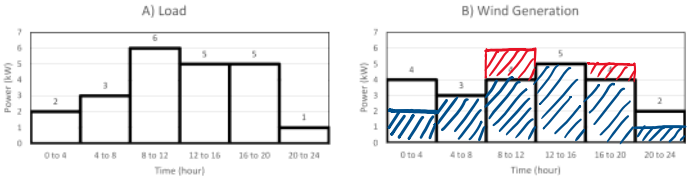
https://ul.org/research-updates/what-are-lithium-ion-batteries/

- A more recent technology, these batteries charge and discharge by the movement of lithium ions between the electrodes.
- Originally found popularity in the portable consumer and business electronic products market; advances in product safety and overall performance are now being driven by their use in the automotive electric vehicle market.
- Li-Ion batteries need protection systems to prevent thermal runaway.
- **Advantages**
  - Lighter, having between 2 and 5 times better storage capacity in terms of weight of the battery than lead-acid types.
  - Between 10 and 20 times more charge-discharge cycles than lead-acid types, making them more suitable for frequent charge-discharge uses associated with load profile shifting and domestic generation technologies such as solar PV.
  - Operational life similar to VRLA.
- **Disadvantages**
  - Without protection systems, can be subject to thermal runaway.
  - Not easily recyclable at present.
  - Mechanical damage may lead to thermal runaway if not adequately protected against.

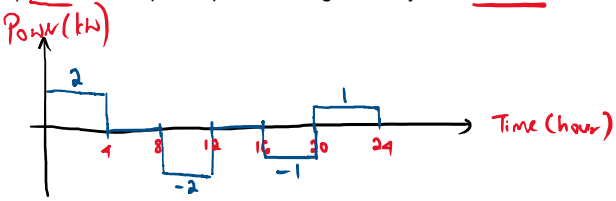
Example

Sunday, October 18, 2026 11:16 PM

A new small community plan to build an off grid energy system. The load and wind generation are given as below.



a) Draw the net power profile through the day and calculate the total net energy per day



Energy = Pt = (2×4) + (-2×4) + (-1×4) + (1×4)
= 0 kWh

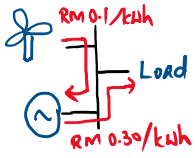
b) A 2kWh battery can be purchased for RM 500. Determine how many batteries are needed to ensure the load can be achieved without a connect on to the grid and what is the cost.

Time	Store (kWh)	Drain (kWh)
0-4	8	0
4-8	0	0
8-12	0	8
12-16	0	0
16-20	0	4
20-24	4	0

A 12 kWh is required to meet the load requirement without connection to the grid.

6 × 2 kWh or 6× RM 500 = RM 3,000

c) Instead of using batteries, a grid connection is suggested. This connection is offered at no cost. The unit buy price is RM0.3/kWh and the sell price is RM0.1/kWh. Calculate the net energy cost per day.



Time	Sell (kWh)	Buy (kWh)	Cost: Sell (RM)	Cost: Buy (RM)	Net (RM) (Sell)
0-4	8	0	0.1 x-8	0	-0.8
4-8	0	0	0	0	0
8-12	0	8	0	0.3 x 8	2.4
12-16	0	0	0	0	0
16-20	0	4	0	0.3 x 4	1.2
20-24	4	0	0.1 x -4	0	-0.4

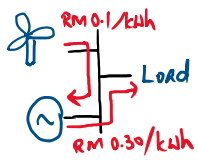
Total = 2.4 per day

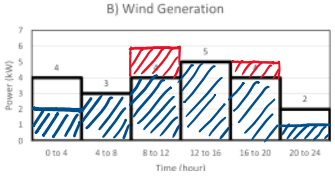
RM 2.4 per day

payback = investment cost / saving cost

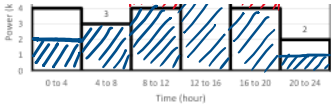
d) If the wind turbine and auxiliary equipment cost RM20,000 to install, calculate the buyback me on:

i. The grid connected option





Time	Sell (kWh)	Buy (kWh)	(From wind) Own consumption	Cost: Sell (RM)	Cost: Buy (RM)	otherwise from grid: Saving	Net (RM)
0-4	8	0	8	0.1 x-8	0	0.3 x -8	-3.2
4-8	0	0	12	0	0	0.3 x -12	-3.6
8-12	0	8	16	0	0.3 x 8	0.3 x -16	-2.4
12-16	0	0	20	0	0	0.3 x -20	-6
16-20	0	4	16	0	0.3 x 4	0.3 x -16	-3.6
20-24	4	0	4	0.1 x -4	0	0.3 x -4	-0.8



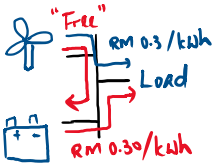
16-20	0	4	16	0	0.3×4	0.3×-16	-3.6
20-24	4	0	4	0.1×-4	0	0.3×-4	-0.8

Total: -19.6 per day

$$\text{Buyback} = \frac{\text{Total investment cost}}{\text{Total saving}} = \frac{20{,}000}{19.6} = 1020 \text{ days}$$

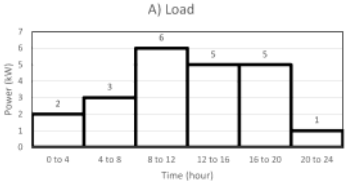
≈ 2.8 years

ii. The battery option



$$\text{Buyback} = \frac{(\text{wind}) \quad 20{,}000 + (\text{battery}) \quad 3{,}000}{26.4} = 871 \text{ days}$$

≈ 2.4 years



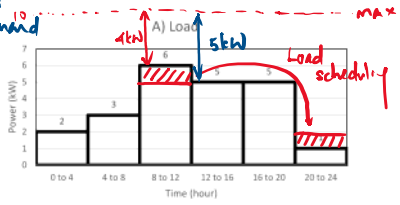
$$\text{Saving} = [(2\times4) + (3\times4) + (6\times4) + (5\times4) + (5\times4) + (1\times4)] \times 0.3$$

$= \text{RM } 26.4 / \text{day}$

e) Suggest a method other than batteries or grid connection that could be used to ensure the communities load is always met.

i) Peak shaving → discharge BESS during peak demand

ii) Load scheduling



10kW max from grid per day

↳ TOU